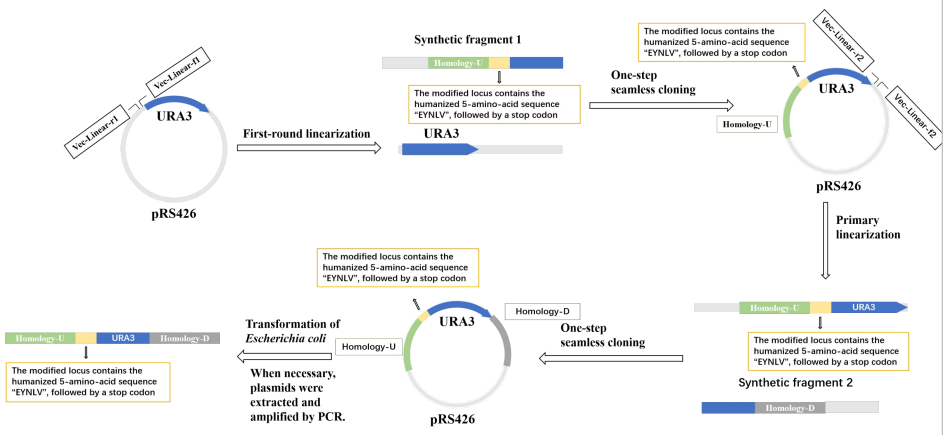
Experimental Record on Humanization Engineering of *GPA1* in *Saccharomyces cerevisiae*

No 1 Date：2026.04.12-04.15 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	C-terminal humanization engineering of the *GPA1* gene (obtaining the target fragment via engineering of the *pRS426* plasmid): design of primers, homology arms, and the Gibson assembly strategy
Note	YeastGenome（https://www.yeastgenome.org/） NCBI（https://www.ncbi.nlm.nih.gov/） SnapGene（https://www.snapgene.com/） Addgene（https://www.addgene.org/）
Result	 **Basic workflow**

Forward primer	Reverse primer	Objective
Vec-Linear-f1:AGAGTGCACCATACCACAGC	Vec-Linear-r1:GGAGCTGCATGTGTCAGAGG	Initial linearization upstream of *URA3*
Val-Hom-U-f:CCCAGGCATTACAGAAACCG	Val-Hom-U-r:TCGTTCTTCCTTCTGTTCGG	Verification of whether the upstream homology arm has been introduced into the plasmid
Vec-Linear-f2:GACCGAGATAGGGTTGAGTGT	Vec-Linear-r2:CCGATTTCGGCCTATTGGTT	Secondary linearization downstream of *URA3*
Val-Hom-D-f:AACCAATAGGCCGAAATCGG	Val-Hom-D-r:CTCGGTCGCCTAGTAGATCT	Verification of whether the downstream homology arm has been introduced into the plasmid
Donor-f:CCCAGGCA	Donor-r:CTCGGTCG	Final linearization,

	TTACAGAAACCG	CCTAGTAGATCT	followed by subsequent transformation into *BY4741*
	Val- Yeast- f:AGAGTGCACCATACCACAGC	Val- Yeast- r:CCGATTTCGGCCTATTGGTT	Verification of successful transformation and homologous recombination within yeast

Primer design

TTCGGTGATGACGGTGAAAACCTCTGACACATGCAGCTCCCAGGCATTAC
AGAAACCGAATTTAACATCGGCTCGTCCAAATTCAAGGTTCTCGACGCTG
GTGGGCAGCGTTCTGAACGTAAGAAGTGGATTCATTGTTTCGAAGGAATT
ACAGCAGTTTTATTTGTTTTAGCAATGAGTGAATACGACCAGATGTTGTTT
GAGGATGAAAGAGTGAACAGAATGCATGAATCAATAATGCTATTTGACAC
GTTATTGAACTCTAAGTGGTTCAAAGATACACCGTTTATTTTGTTTTTAAA
TAAAATTGATTTGTTCGAGGAAAAGGTAAAAAGCATGCCCATAAGAAAGT
ACTTTCCTGATTACCAGGGACGTGTCGGCGATGCAGAAGCGGGTCTAAAA
TATTTTGAGAAGATATTTTTGAGCTTGAATAAGACAAACAAACCAATCTA
CGTGAAACGAACCTGCGCTACCGATACCCAAACTATGAAGTTCGTATTGA
GTGCAGTCACCGATCTAATCATCCAGCAAAACCTTAAAGAATATAATTTG
GTTTGA（EYNLV）
AGAGTGCACCATACCACAGCTTTTCAATTCAATTCATCAT

Synthetic upstream homology arm carrying homology sequences to the linearized plasmid

GTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGAGGAACTGTA
TAATTAAAGTAGTGTTTAGATACGTAAATTCTGTTTCCGAAGATGCAAGA
AGGAGCAGCAGCACCAGAAAAAATTACTATTTTTCTTCTCCATTAGAGTCT
ATGATGGAATGCCAAATGAAAAAGCCATTTTGTTCAACAGTTCTTGATCTC
GTTAAATCGTTCCGGGTTTTCAATTGAAAAACAAGGGTAATAAAATCGCA
TGAGAAAAAAAAGGTCCAGACACTTTTTTAGAAGAAAAACTCTTGAAAAC
AATTGATTCTTCTACGTAAGGACAGGGTATTATCTCGAATTCTTCTCCCTG
GTAAATACTCGGGATATATAGAACCTATTTTGCATCCTATTCTCACTTTTC
TTTATTGTTGTTAATTTCTGGACATTGTTTCATTAATTGAATGAATGCCCTA
AAATATTTCTCAAATCATTTAATAACTACAAAGAAACAAAAAAAAATCAA
TGTTGAGGTGACAAAGAATCAAGATCTACTAGGCGACCGAGATAGGGTTG
AGTGTTGTTCCAGTTTGGAACAAG

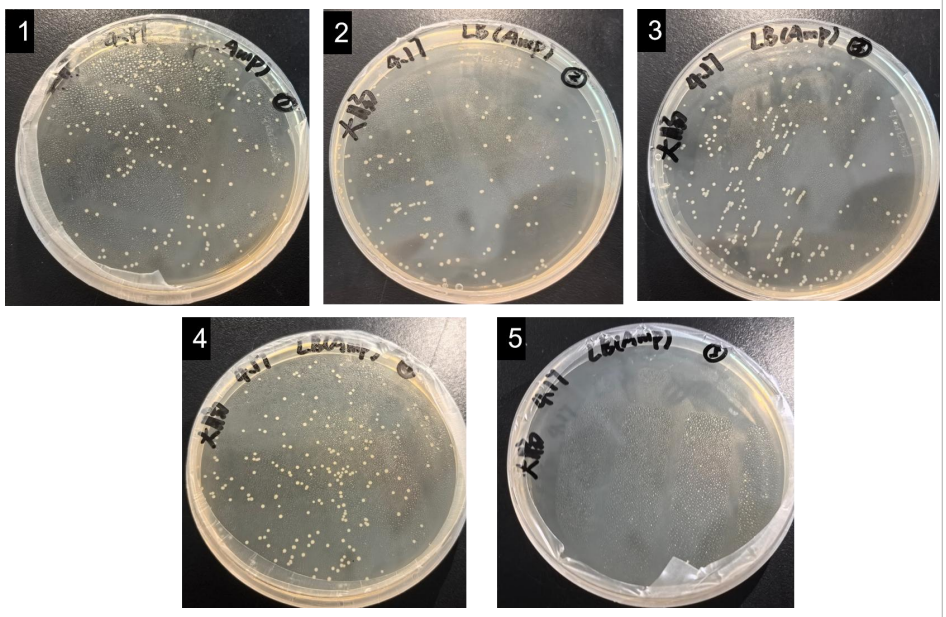
Synthetic downstream homology arm carrying homology sequences to the linearized plasmid

Explanation	This experiment completed the design of the full set of primers and homology arm sequences for the C-terminal humanization engineering of the *GPA1* gene.

	The five C-terminal amino acids of yeast *GPA1*, KIGII, were replaced with EYNLV of human Gαq, and a TGA stop codon was added, with *URA3* selected as the selection marker. Because the GC content of the homology arm regions was excessively low, direct PCR amplification was difficult; therefore, artificially synthesized upstream and downstream fragments carrying plasmid homology overlapping sequences were employed, and vector construction was accomplished through two rounds of Gibson Assembly. Plasmid verification primers for each step and genome identification primers for post-yeast-transformation analysis were designed, thereby laying the foundation for subsequent vector construction and yeast transformation experiments.	
Next plan	Synthesize primers and upstream/downstream homology arm fragments, and conduct wet-lab experiments for vector construction and yeast transformation.	

No 2 Date：2026.04.17- 04.25 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Transformation of the *pRS426* plasmid into *Top10* competent cells, bacterial culture expansion, plasmid extraction, and double restriction enzyme digestion verification.
Note	Protocol 1(E. coli Transformation and Culture Expansion)、Protocol 2(Glycerol Stock Preparation and Cryopreservation)、Protocol 3(Plasmid Extraction from E. coli)、Protocol 4(Restriction Enzyme Digestion)、Protocol 5(Agarose Gel Electrophoresis)
Result	4.19



Transformation results

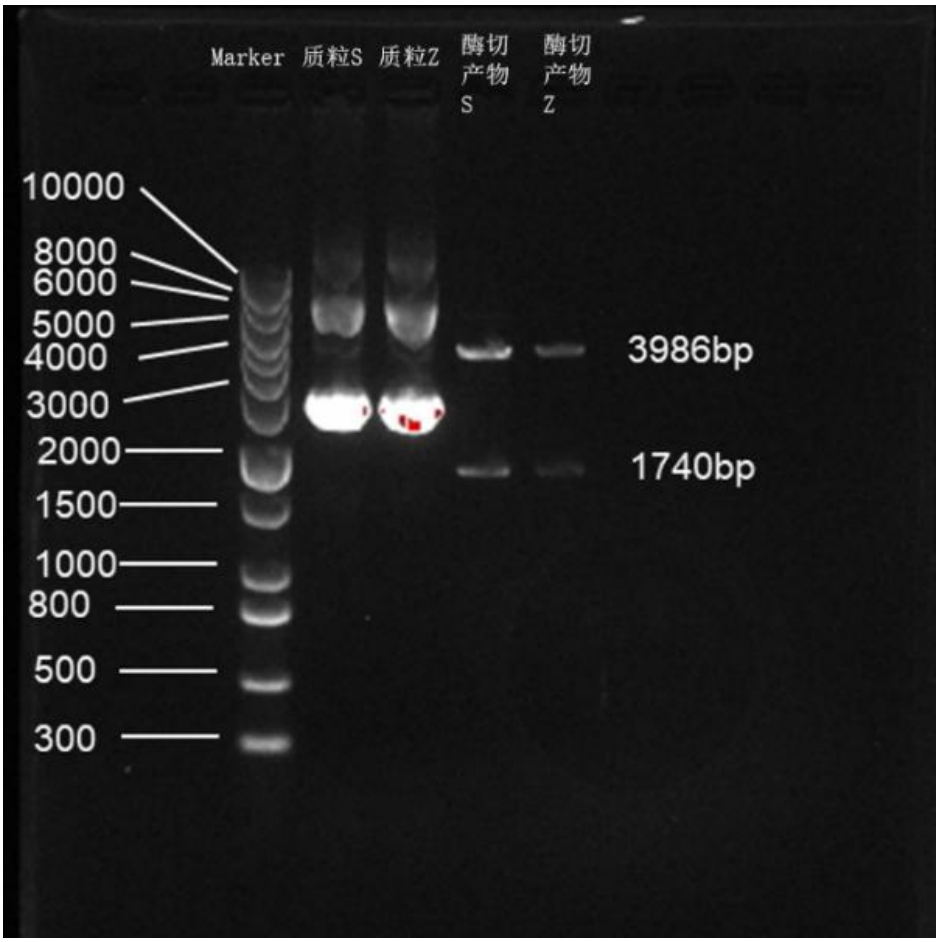
1–4 transformed with the *pRS426* plasmid, 5 blank control

Sample	Concentration (ng/μL)	A260/A280
1	229.30	1.802

2	287.60	1.905
3	294.10	1.907
4	259.75	1.899
5	282.50	1.903
6	285.60	1.901

Concentration and purity of the extracted *pRS426* plasmid

4.25



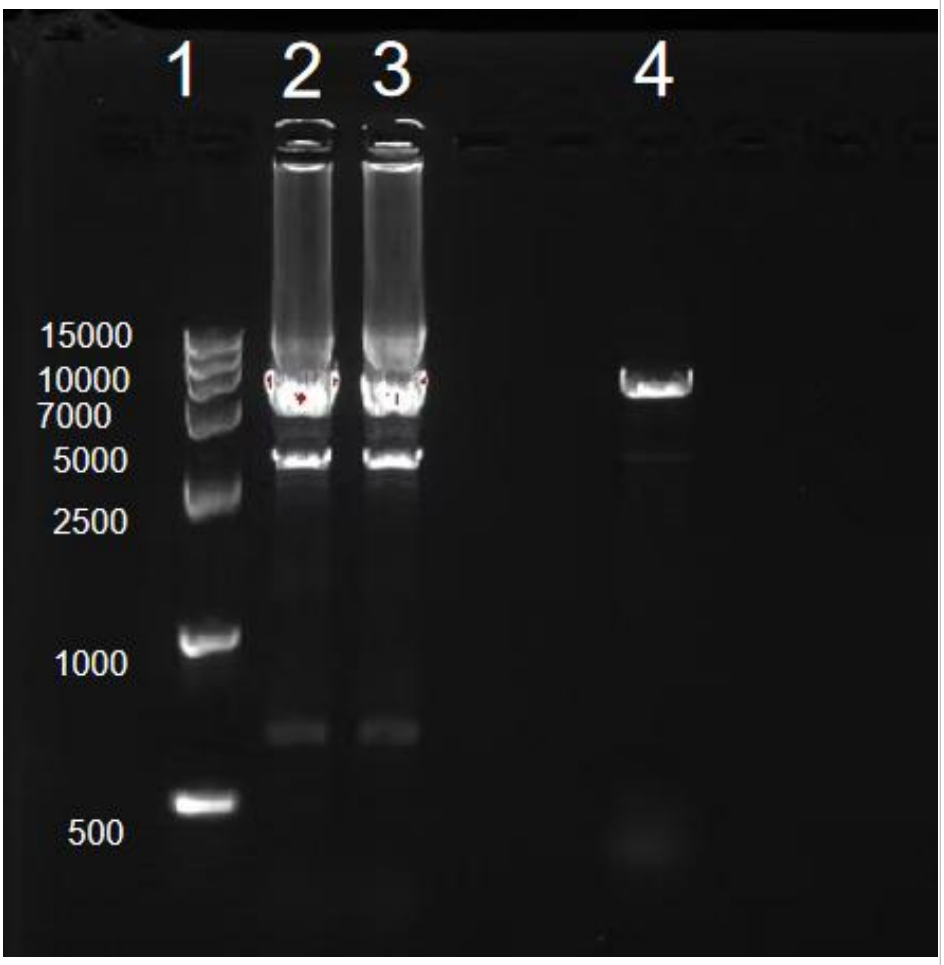
Double restriction enzyme digestion verification (the target bands are 1740 bp and 3986 bp, respectively)

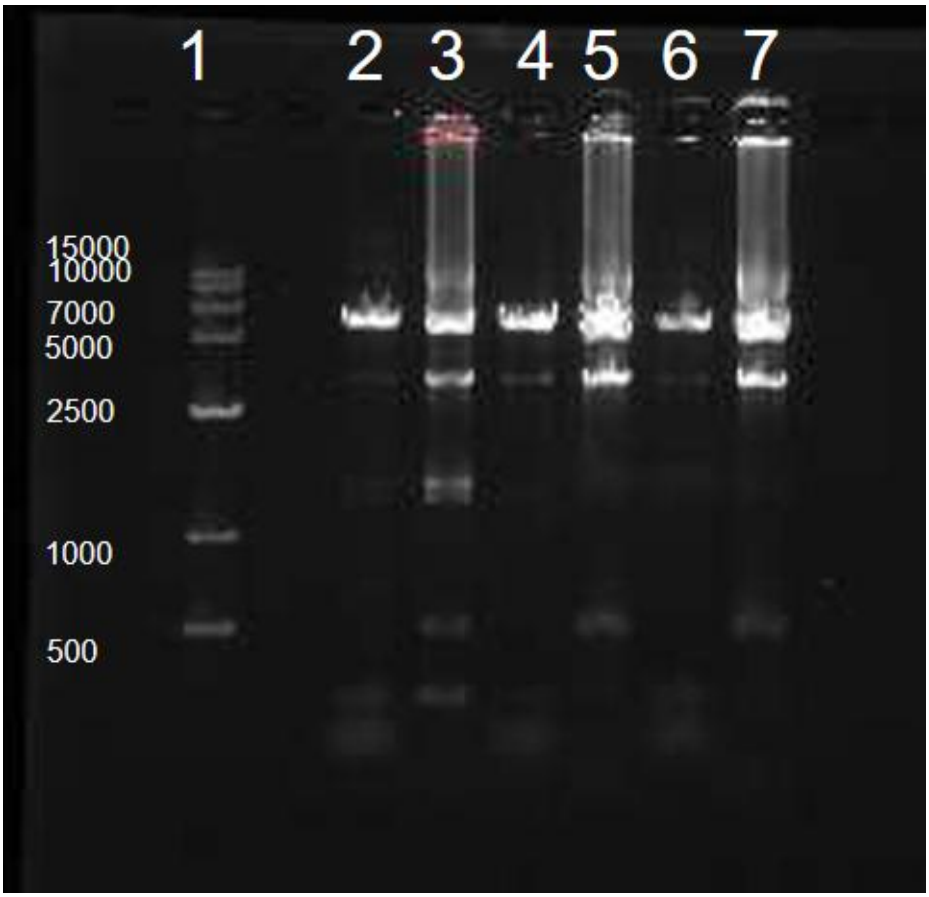
Explanation

This experiment completed the *Escherichia coli* transformation of the *pRS426* plasmid, expansion culture of positive clones, plasmid extraction, and double restriction enzyme digestion identification. During the initial plasmid extraction, residual ethanol on the column membrane caused inhibition of restriction enzyme digestion, and the plasmid yield from small-volume shaking culture in test tubes was relatively low. Subsequent procedural optimization was performed: the empty column was sufficiently centrifuged and air-dried to remove ethanol, culture expansion was switched to 100 mL Erlenmeyer flasks, and the restriction digestion system was prepared sequentially on ice, thereby successfully obtaining high-purity *pRS426* plasmid with correct structure, which can serve as the backbone vector for subsequent Gibson Assembly construction of the *GPA1*

	humanization-engineered plasmid.
Next plan	Upon arrival of the DNA fragments to be synthesized, PCR linearization of the vector and Gibson Assembly experiments were carried out.

No 3 Date：2026.05.10- 05.25 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

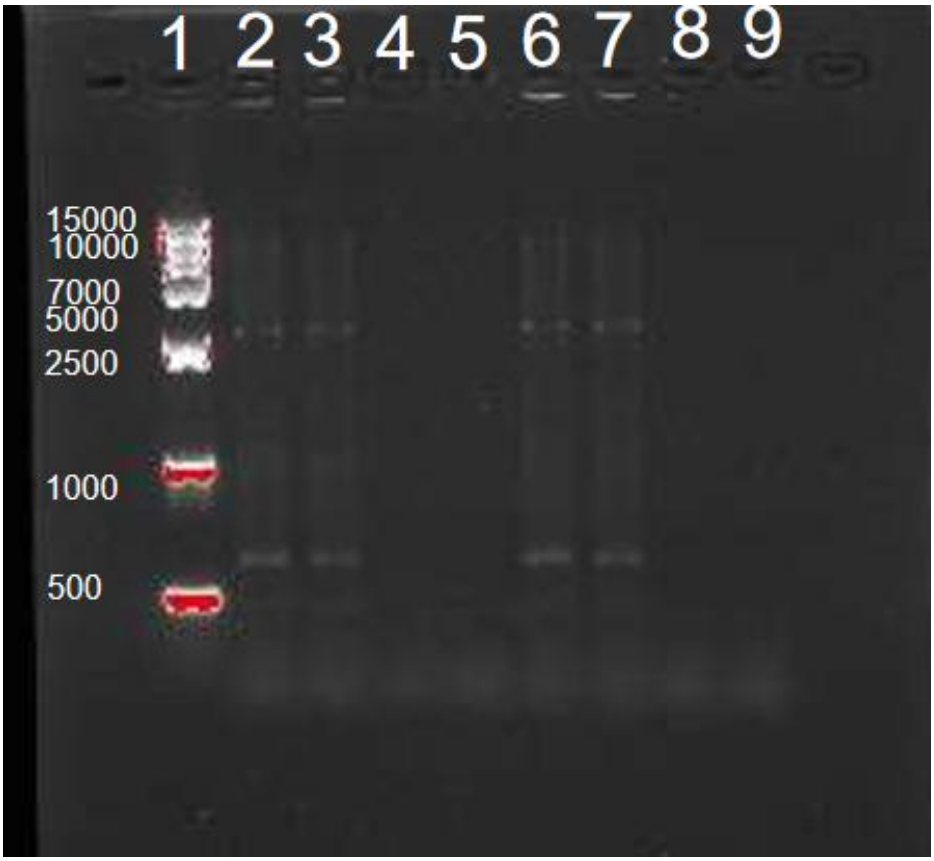
Experiment	Initial PCR linearization of the *pRS426* plasmid, electrophoresis verification, and gel extraction recovery.
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)
Result	5.10  **First preliminary experiment for initial plasmid linearization (the target band is 6092 bp)** 1: 15000 bp marker; 2–3: experimental group; 4: blank control 5.10



Second preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control (annealing temperature 58°C, template amount 50 ng); 3: experimental group (annealing temperature 58°C, template amount 50 ng); 4: blank control (annealing temperature 60°C, template amount 50 ng); 5: experimental group (annealing temperature 60°C, template amount 50 ng); 6: blank control (annealing temperature 58°C, template amount 100 ng); 7: experimental group (annealing temperature 58°C, template amount 100 ng)

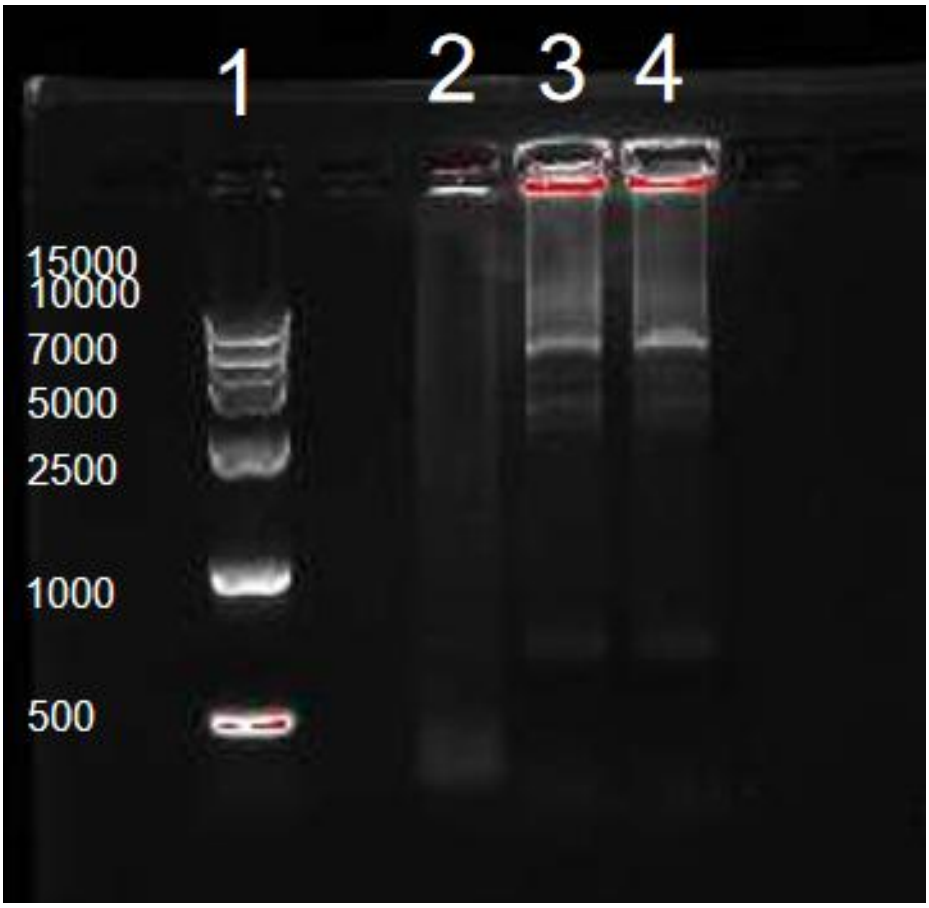
5.11



Third preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2–3: experimental group (annealing temperature 60°C); 4–5: blank control (annealing temperature 60°C); 6–7: experimental group (annealing temperature 62°C); 8–9: blank control (annealing temperature 62°C)

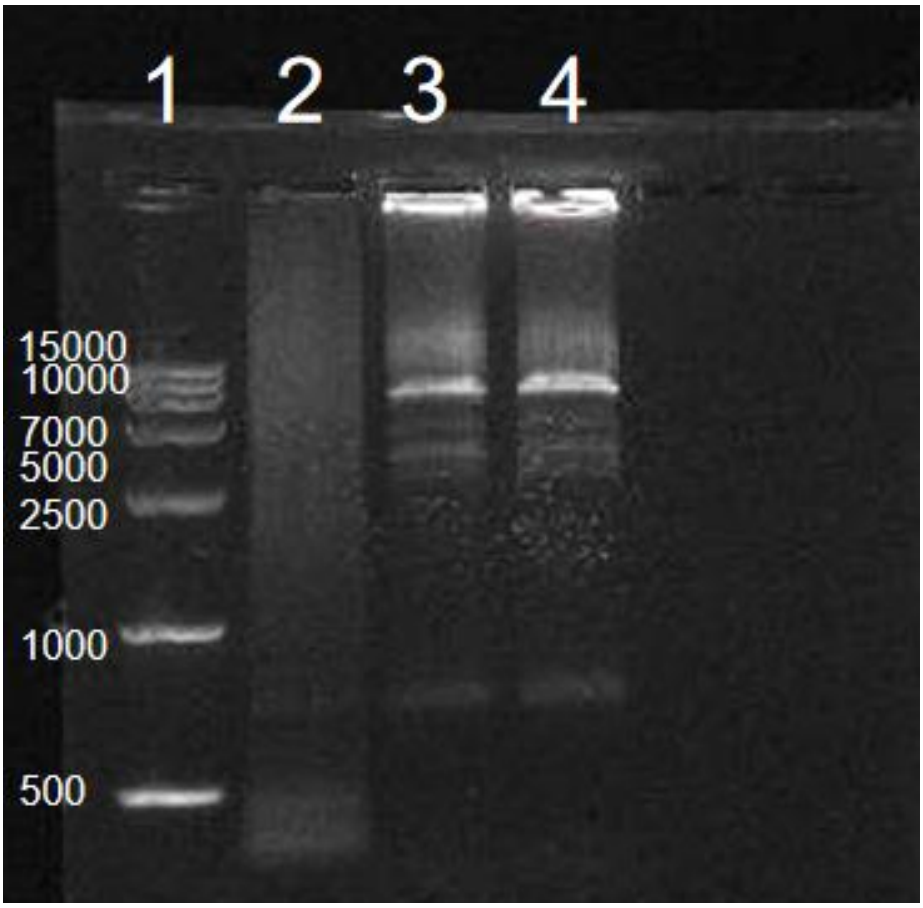
5.12



Fourth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

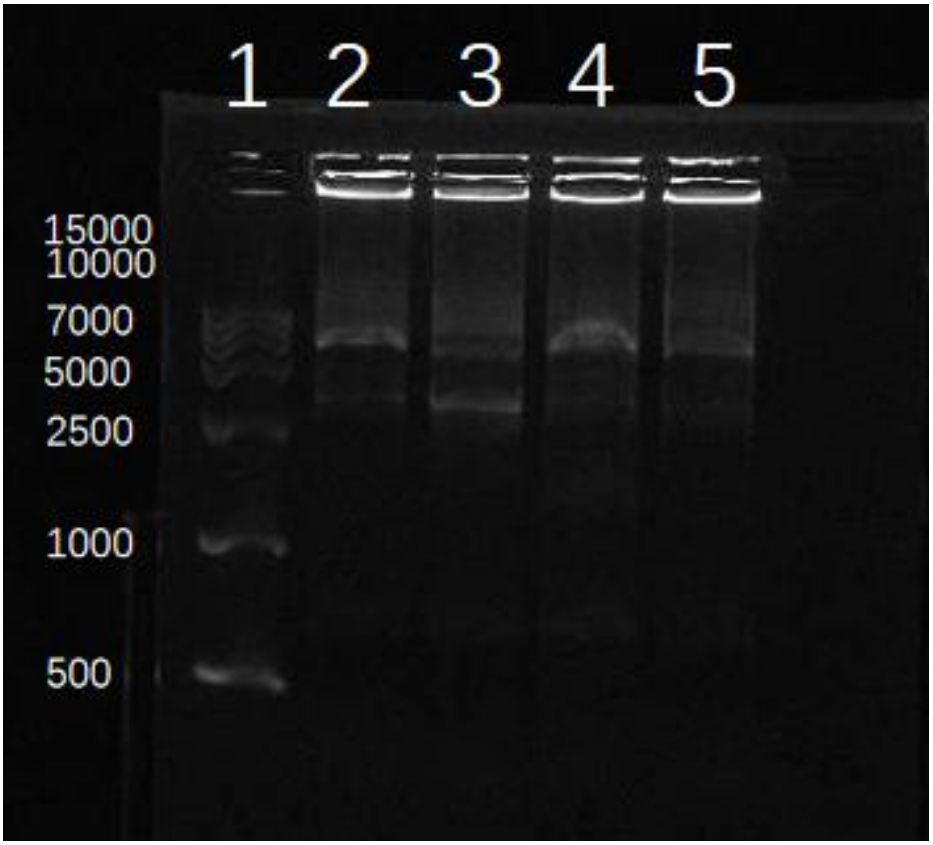
5.12



Fifth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

5.19



Sixth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2–5: experimental group

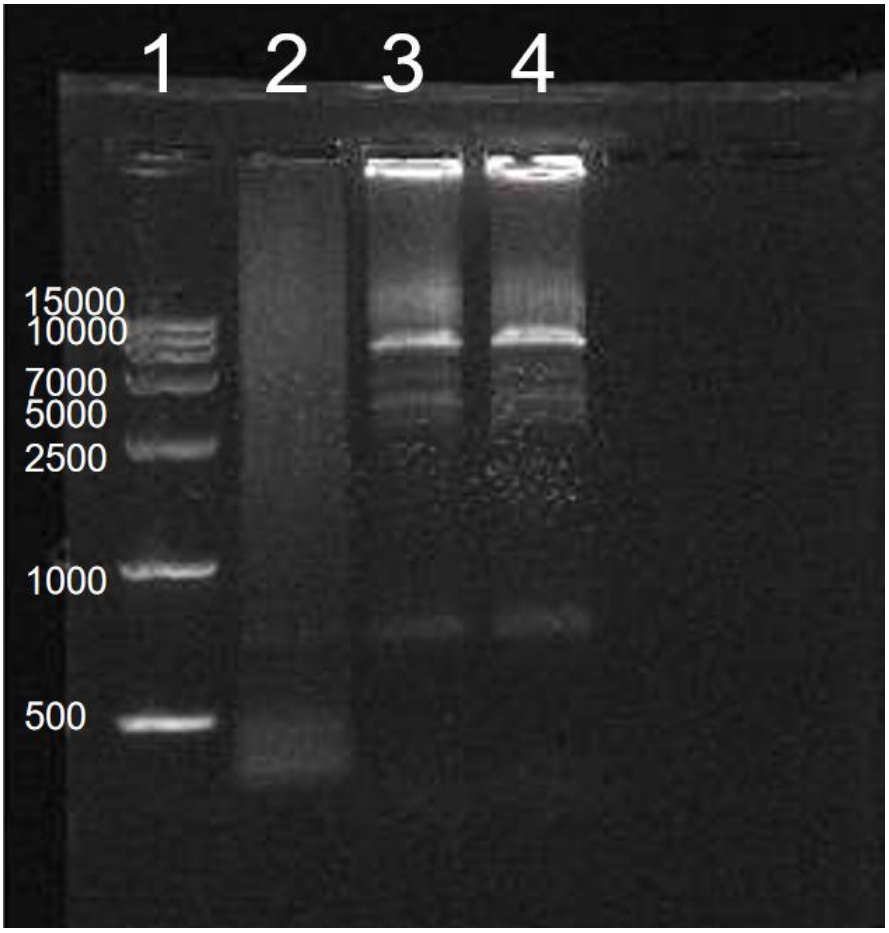
5.23



Seventh preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2–3: experimental group (annealing temperature 60°C); 4–5: blank control (annealing temperature 60°C); 6–7: experimental group (annealing temperature 62°C); 8–9: blank control (annealing temperature 60°C)

5.25



Formal experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

Sample	Concentration (ng/μL)	A260/A280
1	14.5	2.042
2	12.4	2.756

Concentration and purity of the linearized product recovered by gel extraction

Explanation	In this experiment, the *pRS426* plasmid previously verified as correct by double restriction enzyme digestion was used as the template, and PCR was performed with the primers VecLinear_f1/VecLinear_r1 to accomplish the initial linearization of the plasmid (the blank control had consistently shown a band in the earlier stage, and a considerable amount of time was spent excluding contamination sources, ultimately confirming that it was likely caused by operational issues). Agarose gel electrophoresis confirmed that the size of the PCR product band matched the expectation, with no obvious non-specific bands; subsequently, gel extraction recovery was carried out to remove residual circular plasmid template and impurities, yielding the purified linear vector *pRS426_L*, which provides a qualified linear backbone for the Gibson Assembly of the

	upstream homology arm fragment.
Next plan	Gibson Assembly of the upstream homology arm fragment with *pRS426_L* and *Escherichia coli* transformation were carried out.

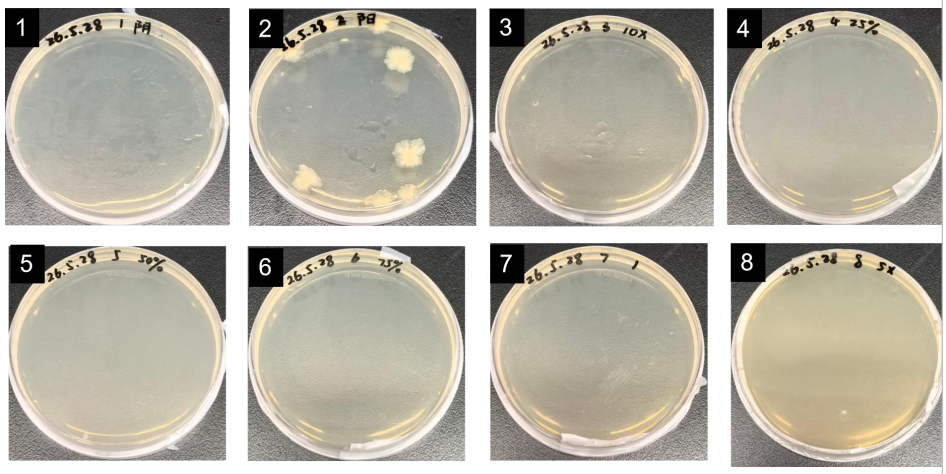
No 4 Date：2026.05.25- 05.30 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	One-step cloning (Gibson Assembly) of the upstream homology arm fragment with *pRS426_L* and *Escherichia coli* transformation.
Note	Protocol 11(Gibson Assembly)、Protocol 1(E. coli Transformation and Culture Expansion)
Result	

Component	Volume
Volume	2.5 μL（50 ng）
Linearized vector	1.5 μL（Approximately 20 ng）
2×CE Mix V3	5 μL
DdH_2O	1 μL

One-step cloning system

5.30



Transformation plate images

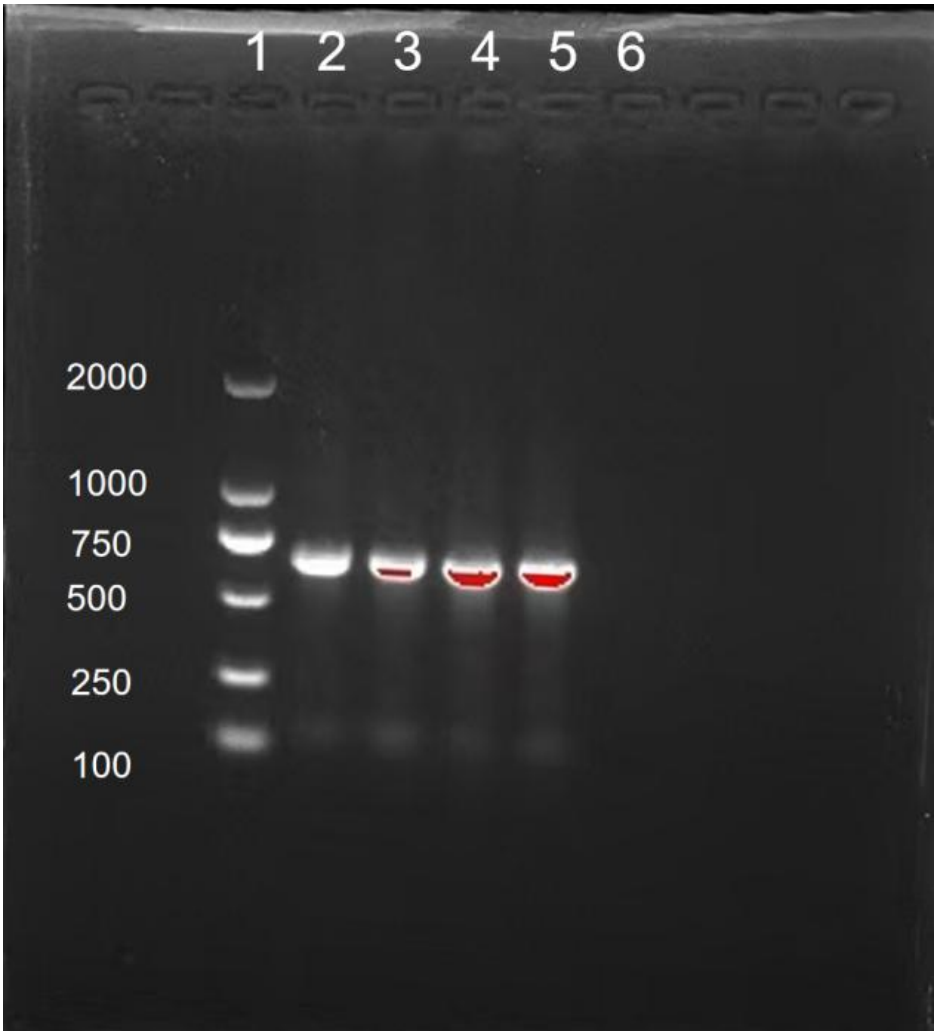
1: blank control; 2: transformed with *pRS426*; 3: one-step cloning product, plated after 10-fold concentration; 4: one-step cloning product, plated after dilution to 25% of the original solution; 5: one-step cloning product, plated after dilution to 50% of the original solution; 6: one-step cloning product, plated after dilution to 75% of the original solution; 7: one-step cloning product, plated as the original solution; 8: one-step cloning product, plated after 5-fold concentration

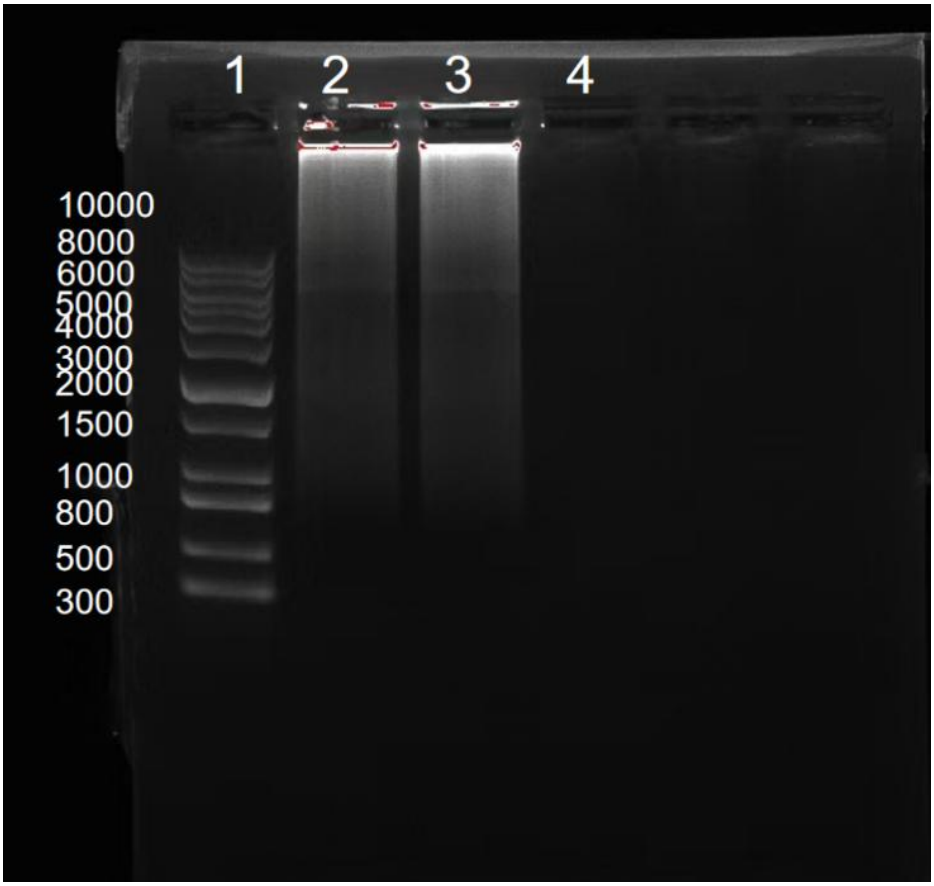
Explanation	In this experiment, the purified linear vector *pRS426_L* was subjected to Gibson one-step cloning assembly with the artificially synthesized upstream homology arm fragment. The blank control excluded background contamination from the plates and competent cells; the positive control exhibited bacterial contamination and thus lost its reference significance; multiple concentration gradient plating was set up, yet *Escherichia coli* positive transformants could not be obtained throughout. PCR verification demonstrated that the *in vitro* system had completed

	correct fragment assembly, but the assembled product was unable to undergo transformation screening *in vivo* in *Top10*. To avoid repeatedly consuming reagents and time, the experimental strategy was adjusted: obtaining the circular recombinant plasmid *pRS426_circ1 in vivo* in *Escherichia coli* was abandoned, and the one-step cloning reaction mixture was directly used as the template for subsequent secondary PCR linearization of the vector, thereby continuing to advance the gene editing experiment.
Next plan	PCR verification was performed on the one-step cloning reaction stock solution, and the stock solution was used as the template for the secondary PCR linearization experiment of the vector.

No 5 Date：2026.06.01-06.07 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	PCR verification of the one-step cloning reaction stock solution and secondary linearization of the vector
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)

Result	6.1  **Initial one-step cloning electrophoresis verification (the target band is 640 bp)** 1: 2000 bp marker; 2–5: experimental group; 6: blank control 6.7



Electrophoretogram of secondary linearization of the vector (the target band is 6092 bp)

1: 10000 bp marker; 2–3: experimental group; 4: blank control

Sample	Concentration (ng/μL)	A260/A280
1	4.9500	1.623

Concentration and purity of the linearized product recovered by gel extraction

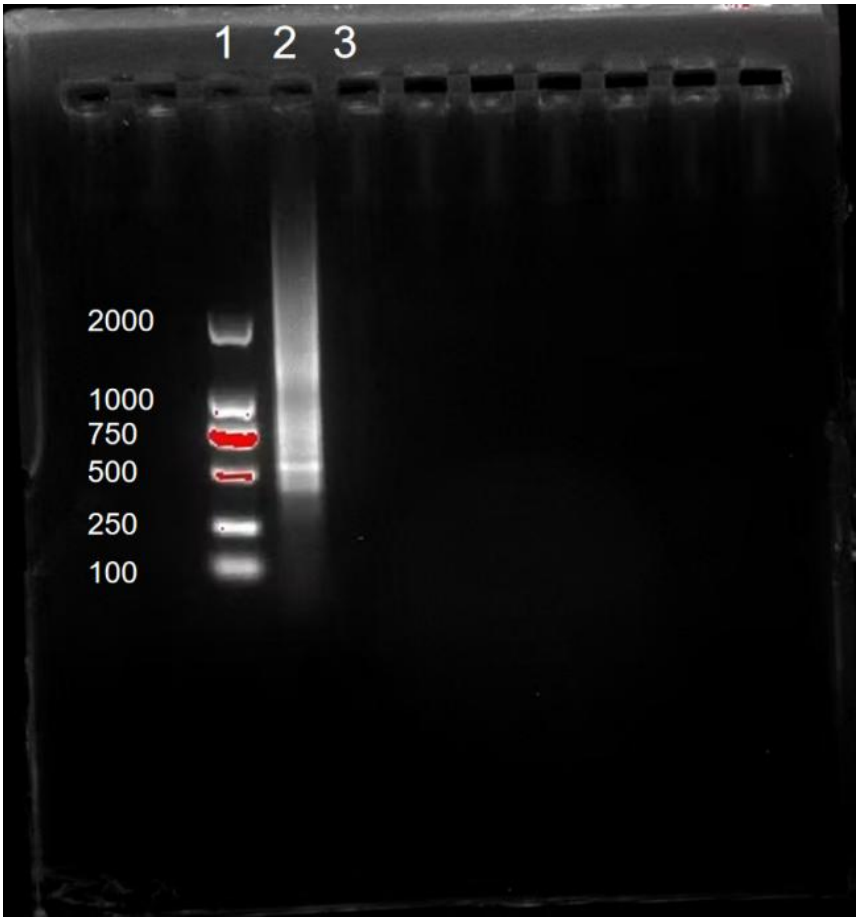
Explanation	Because the circular *pRS426_circ1* plasmid could not be obtained through *Escherichia coli* transformation, the one-step cloning reaction stock solution was directly used as the PCR template, and the expected 640 bp band was amplified, further confirming that the *in vitro* Gibson Assembly had achieved correct splicing of the upstream fragment. The one-step cloning reaction stock solution was used to accomplish the secondary PCR linearization of the vector, and gel extraction recovery yielded the purified *pRS426_L2* linear backbone, thereby circumventing the experimental obstacle caused by the failure of *Escherichia coli* transformation screening and rendering it suitable for the Gibson Assembly of the downstream homology arm fragment.
Next plan	The artificially synthesized downstream homology arm fragment was subjected to Gibson Assembly with the *pRS426_L2* backbone, followed by PCR verification.

No 6 Date：2026.06.07-06.08 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Gibson Assembly of the downstream homology arm fragment with the *pRS426_L2* backbone and PCR verification
Note	Protocol 11(Gibson Assembly)、Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)
Result	6.8

Component	Volume
Linearized fragment(4.9500 ng/μL)	2.5
Downstream homology arm(20 ng/μL)	1.25
2×CE Mix V3	5
ddH_2O	1.25

Secondary one-step cloning system

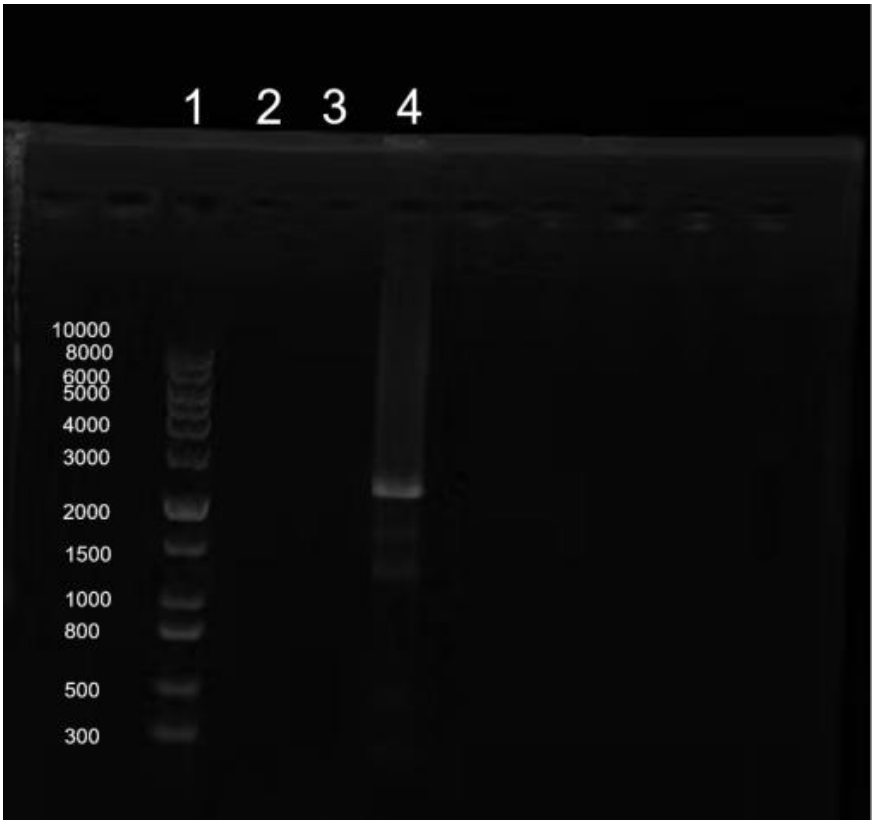


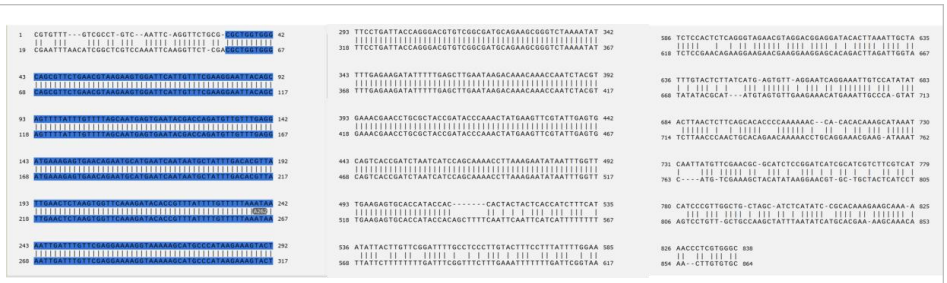
Secondary one-step cloning electrophoresis verification (the target band is 640 bp)

1: 2000 bp marker; 2: experimental group; 3: blank control

Explanation	In this experiment, the *pRS426_L2* linear backbone was subjected to *in vitro* Gibson Assembly with the artificially synthesized downstream homology arm fragment. The assembly stock solution was directly used as the template for PCR verification, and the expected target band was amplified, demonstrating that the downstream homology arm had been successfully spliced with the vector backbone *in vitro*. Combined with the verification results of the upstream fragment from the previous step, this indicates that the complete recombinant sequence comprising the upstream homology arm, the humanized sequence, the *URA3* expression cassette, and the downstream homology arm had already been obtained *in vitro*.
Next plan	Using the Gibson Assembly stock solution from this round as the template, PCR amplification was performed to obtain the complete linearized DNA fragment for yeast transformation.

No 7 Date：2026.06.11-06.15 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	The complete linear target DNA fragment for yeast transformation was amplified and sent for sequencing identification.
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)
Result	6.11  **Obtainment of the target fragment (the target band is 2289 bp)** 1: 12000 bp marker; 2–3: blank control; 3: experimental group 6.15

	 **Comparison of sequencing results** 1: 2000 bp marker; 2: experimental group; 3: blank control
Explanation	Although electrophoresis yielded a linear recombinant DNA fragment of 2289 bp with the expected size, sequencing exposed two critical defects: insufficient primer design for long-fragment sequencing resulted in incomplete sequence coverage; and the accumulation of multiple *in vitro* PCR cycles introduced numerous base mutations, making it impossible to obtain an editing donor fragment with a completely correct sequence. Therefore, the current experimental route combining Gibson Assembly with multiple rounds of PCR amplification was terminated to circumvent the risk of mutations, and the experimental scheme was changed to reconstruct the *GPA1* humanization editing donor fragment using Overlap Extension PCR (fusion PCR).
Next plan	The fusion PCR scheme was adopted to redesign the primers and construct the editing donor fragment.

No 8 Date：2026.06.15-06.18 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

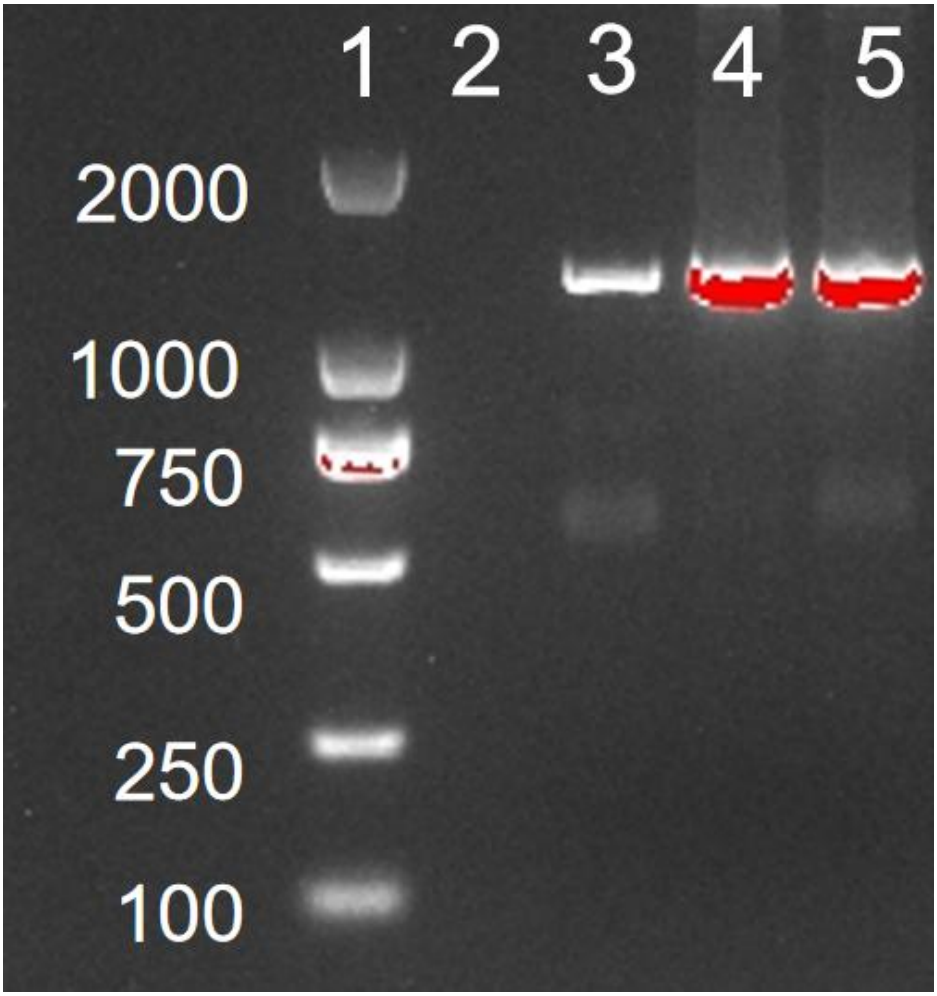
Experiment	Primer design for constructing the *GPA1* humanization donor fragment by fusion PCR (Overlap Extension PCR)
Note	YeastGenome（https://www.yeastgenome.org/） NCBI（https://www.ncbi.nlm.nih.gov/） SnapGene（https://www.snapgene.com/） Addgene（https://www.addgene.org/）
Result	

Forward primer	Reverse primer	Objective
*URA3*_f:AGAGTGCACCATACCACAGC	*URA3*_r:CCGATTTCGGCCTATTGGTT	Obtain *URA3*
U_f:CAGGCATTACAGAAACCGAATTTAACATCG	U_r:ATGATGAATTGAATTGAAAAGCTGTGG	Obtain the upstream homology arm
D_f:GTTAAATCAGCTCATTTTTTAACCAATAGGC	D_r:GCCTAGTAGATCTTGATTCTTTGTCAC	Obtain the downstream homology arm
T1_f:CGGATGAAGACATTTTGAAGGGCC	T1_r:CCGATTTCGGCCTATTGGTT	Verification after
T2_f:CGGATGAAGACATTTTGA	T2_r:CAACAGGACTAGG	transformat

	AGGGCC ATGAGTAGCAGC ion *URA3*_f:AGAGTGCACCATACCA *URA3*_r:CCGATTTCGGCC 获得 *URA3* CAGC TATTGGTT Primer design
Explanation	The earlier Gibson multi-round PCR route was declared a failure due to accumulated mutations from multiple rounds of amplification. In this experiment, based on the fusion PCR scheme in Protocol 8, primer splitting design was carried out for the *GPA1* humanization donor fragment. The complete donor was divided into three segments, A, B, and C, with homologous overlapping sequences added between the fragments, so that the full-length donor could be directly spliced through overlap extension PCR, thereby reducing the number of PCR cycles and lowering the risk of base mutations. The outer amplification primers continued to use the previously verified target DNA_f/r to ensure the correctness of the homology arm sequences at both ends. In this stage, the design of the entire primer strategy was completed, laying the groundwork for the subsequent wet-lab fusion PCR experiments.
Next plan	The complete set of fusion PCR primers was synthesized, and segmented PCR amplification of each DNA fragment was carried out.

No 9 Date：2026.07.06- 07.12 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

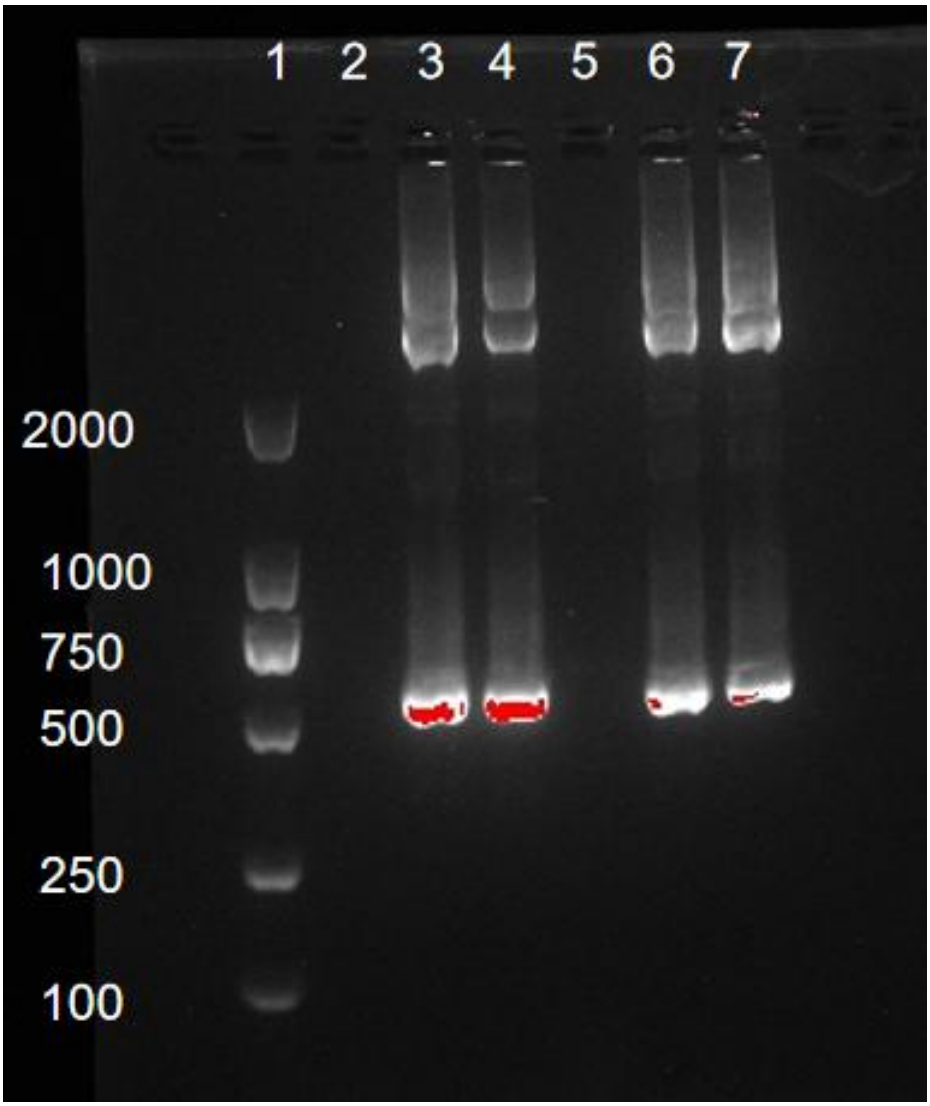
Experiment	Fusion PCR: separate amplification of the upstream homology arm, the *URA3* expression cassette, and the downstream homology arm fragment
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)、Protocol 8（Overlap Extension PCR）
Result	7.7



Preliminary experiment: obtainment of the target fragment (the target band is 1262 bp)

1: 2000 bp marker; 2: blank control; 3–5: experimental group

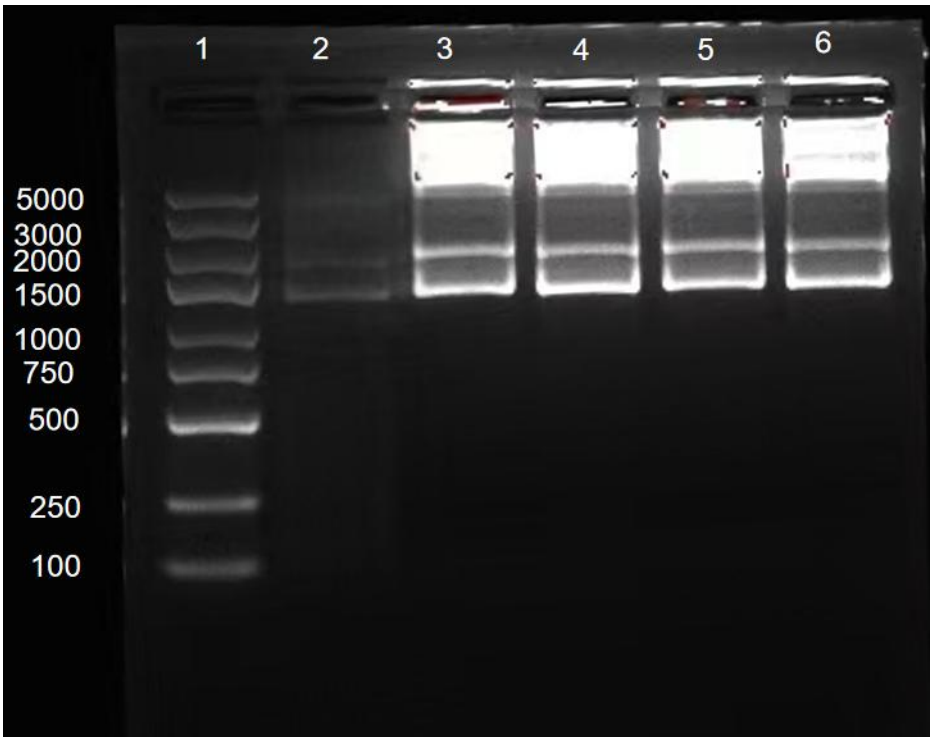
7.9



Preliminary experiment: obtainment of the upstream and downstream homology arms (the target bands are 558 bp and 540 bp, respectively)

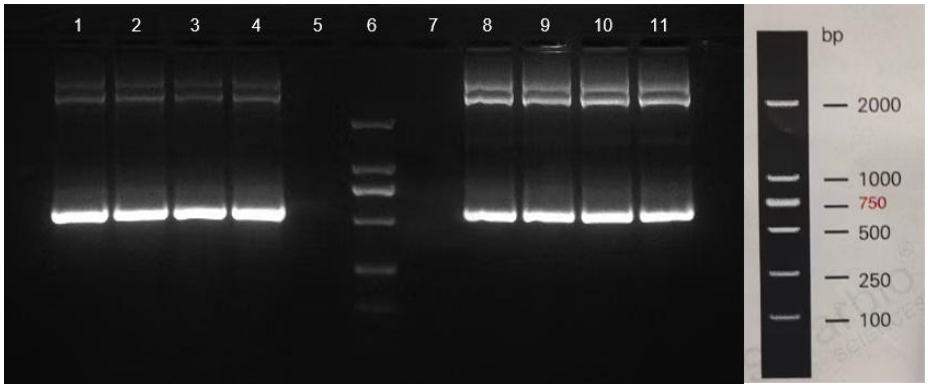
1: 2000 bp marker; 2: blank control for upstream homology arm amplification; 3–4: experimental group for upstream homology arm amplification; 5: blank control for downstream homology arm amplification; 6–7: experimental group for downstream homology arm amplification

7.11



Formal experiment: obtainment of the target fragment (the target band is 1262 bp)

1: 2000 bp marker; 2: blank control; 3–6: experimental group



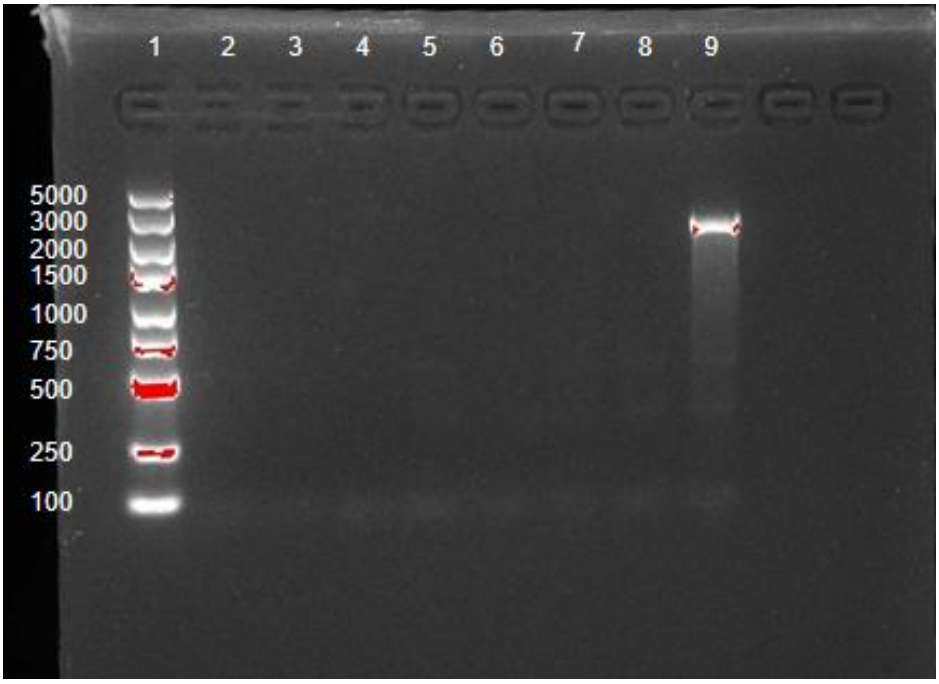
Formal experiment: obtainment of the upstream and downstream homology arms (the target bands are 558 bp and 540 bp, respectively)

1–4: experimental group for upstream homology arm amplification; 5: blank control for upstream homology arm amplification; 6: 2000 bp marker; 7: blank control for downstream homology arm amplification; 8–10: experimental group for downstream homology arm amplification

Sample	Concentration (ng/μL)	A260/A280
Upstream homology arm 1	121	1.814
Upstream homology arm 2	111	1.861

URA3 1	72.95	1.812
URA3 2	140.60	1.668
Downstream homology arm 1	75.45	1.711
Downstream homology arm 2	82.000	1.739

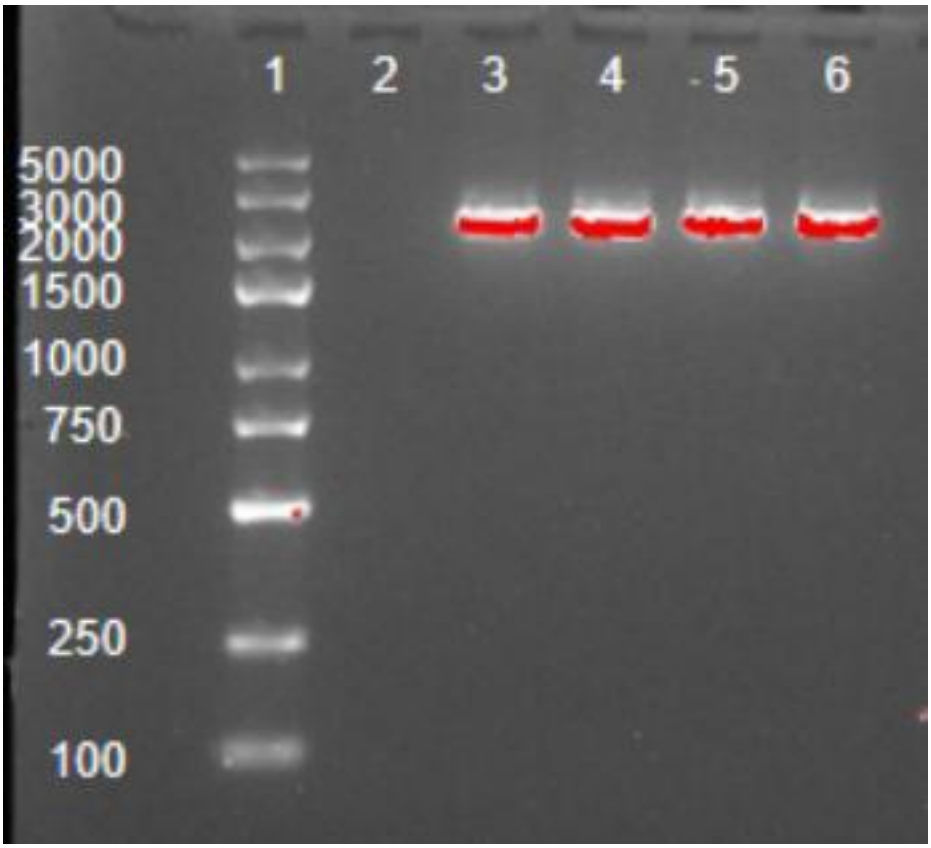
Concentration and purity of the linearized product recovered by gel extraction



Preliminary experiment: fusion PCR (the target band is 2280 bp)

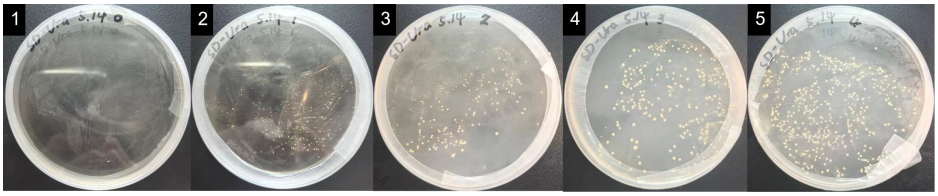
1: 5000 bp marker; 2: blank; 3: upstream homology arm; 3: *URA3*; 4: downstream homology arm; 5: upstream homology arm + *URA3*; 6: *URA3* + downstream homology arm; 7: upstream homology arm + downstream homology arm; 8: upstream homology arm + *URA3* + downstream homology arm

7.12

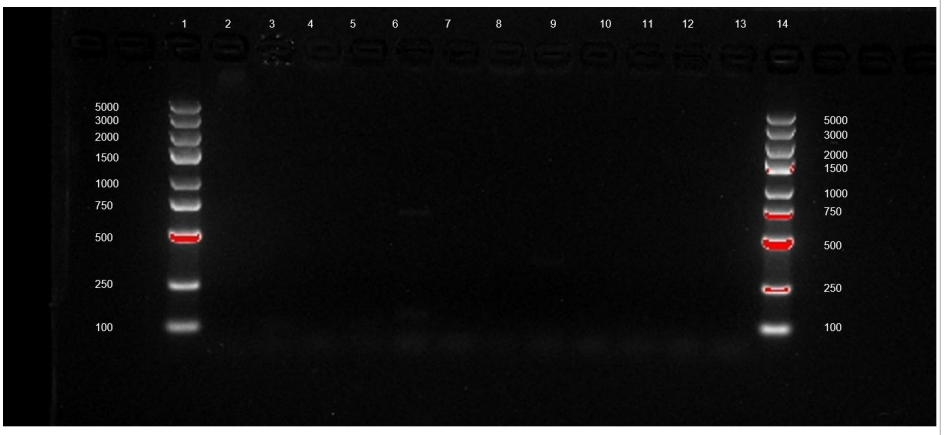
	 **Formal experiment: fusion PCR (the target band is 2280 bp)** 1: 5000 bp marker; 2: blank; 3–6: experimental group
Explanation	This experiment implemented Protocol 8 overlap extension fusion PCR, completing the construction of the full-length donor fragment in two steps. In the first round, the primer-free reaction relied on the homologous overlapping regions between fragments to achieve bridging and splicing of fragments A, B, and C; in the second round, outer primers were added for amplification to obtain the complete 2280 bp editing donor fragment. Compared with the previous multi-round PCR–Gibson route, fusion PCR reduces the number of cycles and lowers the risk of mutations introduced by amplification; the full-length product was recovered and purified.
Next plan	The fusion PCR band was single, and it was prepared for transformation.

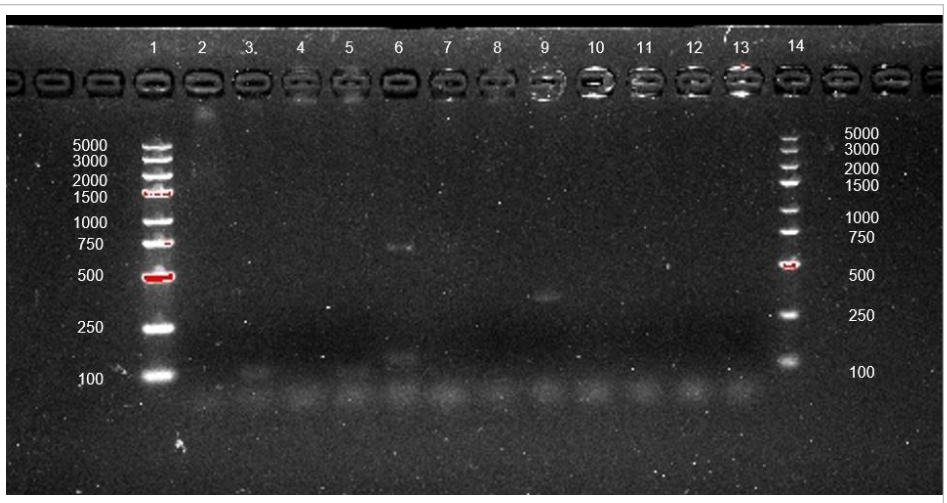
No 10 Date：2026.07.15- 07.18 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	The fusion PCR product was transformed into *Saccharomyces cerevisiae*.
Note	Protocol 12(Yeast Transformation)
Result	7.18

	 **Yeast transformation plate images** 1: blank control; 2–5: experimental group (the date marked on the plates in this figure is the plate-pouring date)
Explanation	The full-length donor fragment obtained by fusion PCR splicing largely circumvented the base mutation problem of the previous Gibson–multi-round PCR scheme. This linear editing donor fragment was introduced into *BY4741* yeast cells, and potential yeast transformants with C-terminal humanization of *GPA1* were obtained on SD-Ura plates by relying on the *URA3* selection marker.
Next plan	Colony PCR verification was performed and the samples were sent for sequencing.

No 11 Date：2026.07.19- 07.23 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

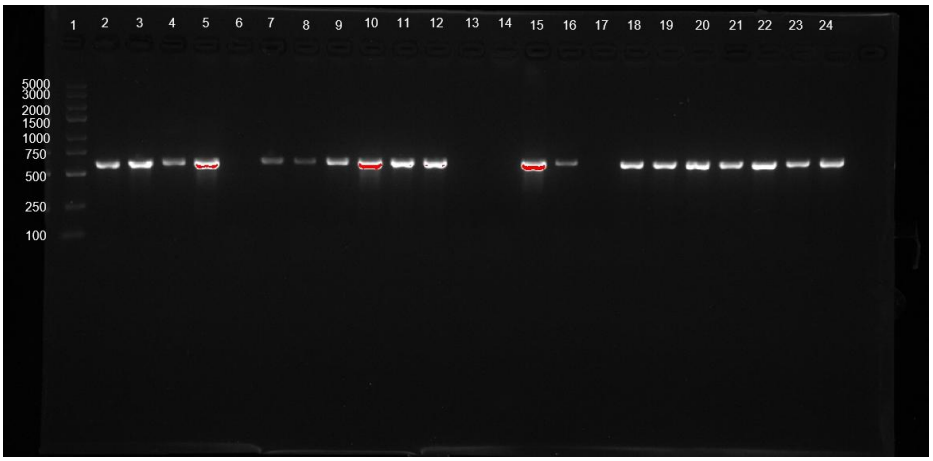
Experiment	Colony PCR verification and analysis of the sequencing results.
Note	Protocol 13(Colony PCR for *Saccharomyces cerevisiae*)、Protocol 5(Agarose Gel Electrophoresis)
Result	7.19  **Electrophoretogram of the first colony PCR (the target band is 2280 bp)** 1: 5000 bp marker; 14: 5000 bp marker; 2–13: corresponding to picked colonies 1–12, respectively



Electrophoretogram of the second colony PCR (the target band is 2280 bp)

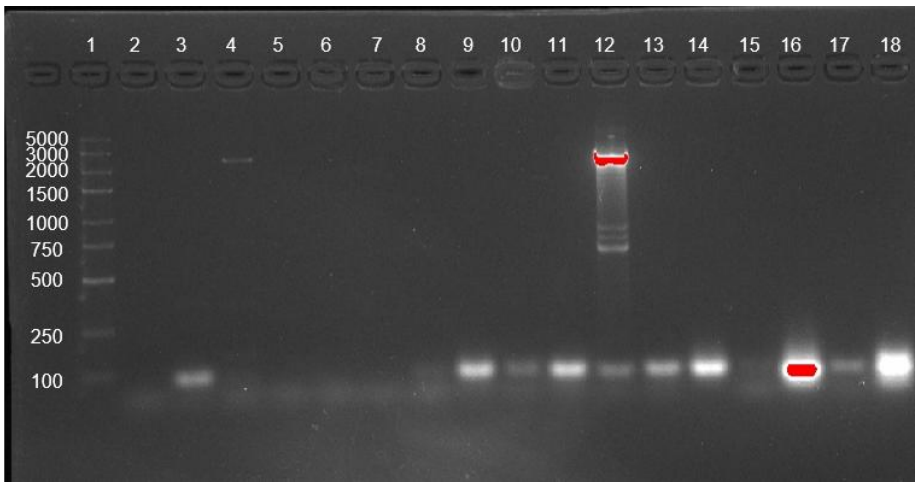
1: 5000 bp marker; 14: 5000 bp marker; 2–13: corresponding to picked colonies 1–12, respectively

7.20



Electrophoretogram of the third colony PCR (verification of the upstream homology arm) (the target band is 558 bp)

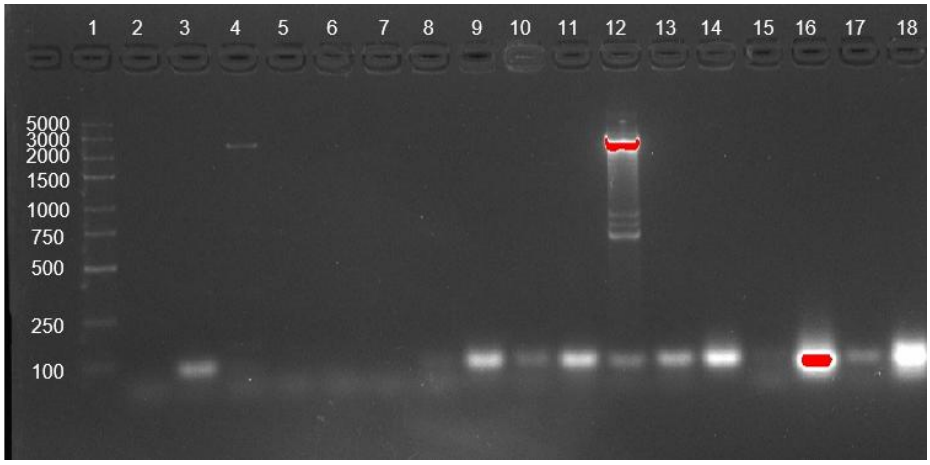
1: 5000 bp marker; 2–24: corresponding to the picked colonies, respectively (1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 8, 5) (because it was felt that 8 and 5 had not been loaded well during sample loading, they were re-loaded)



Electrophoretogram of the fourth colony PCR (verification of the full length) (the target band is 2280 bp)

1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24)

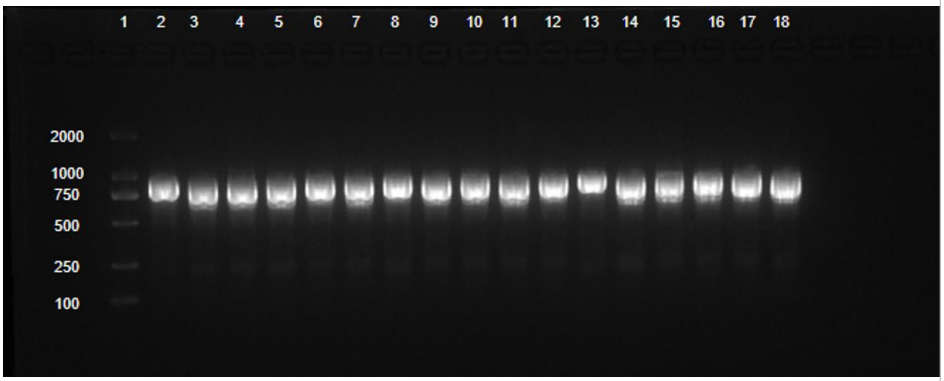
7.21



Electrophoretogram of the fifth colony PCR (verification of the full length) (the target band is 2280 bp)

1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24)

7.22

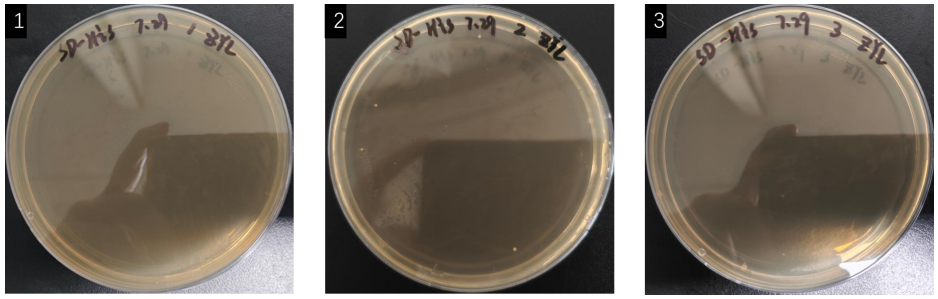
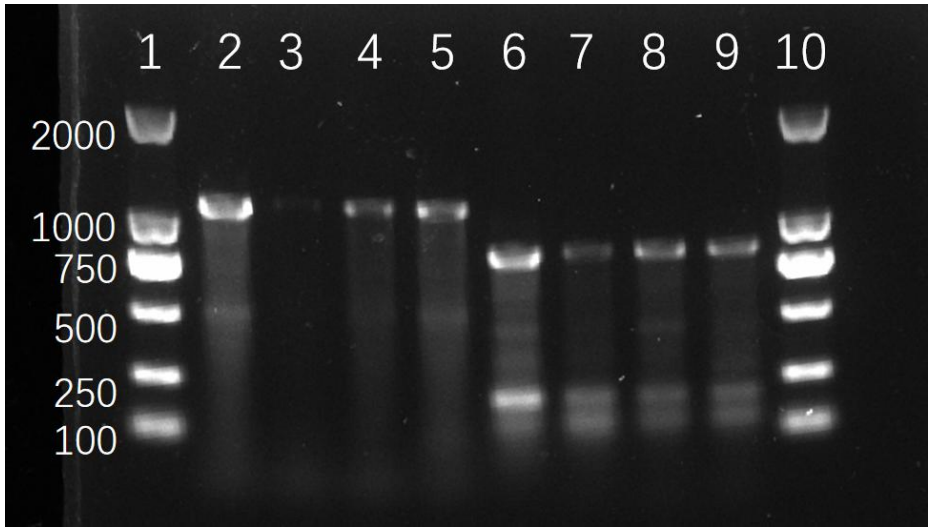


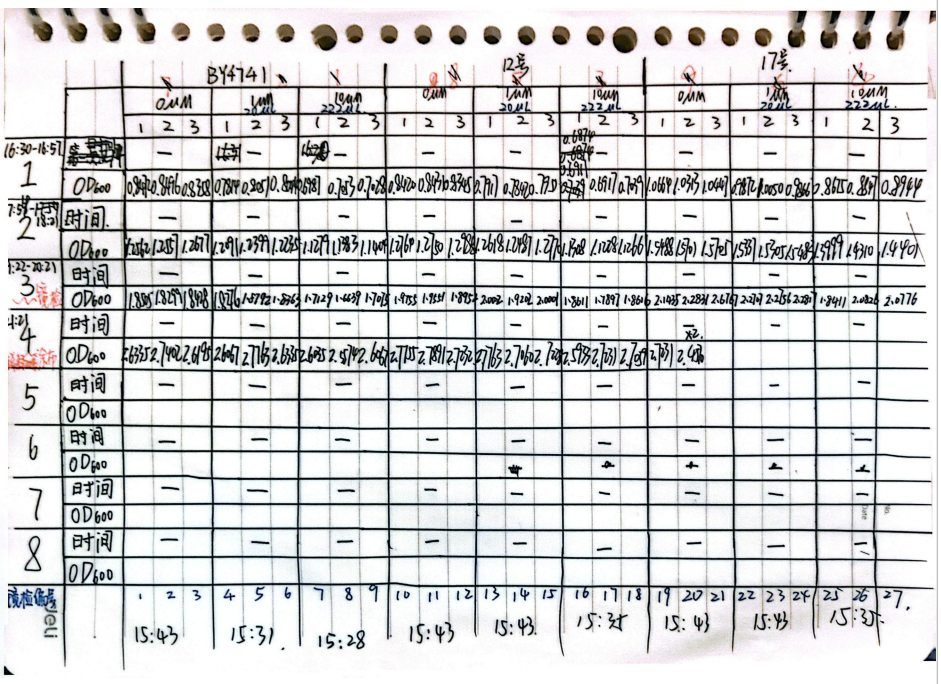
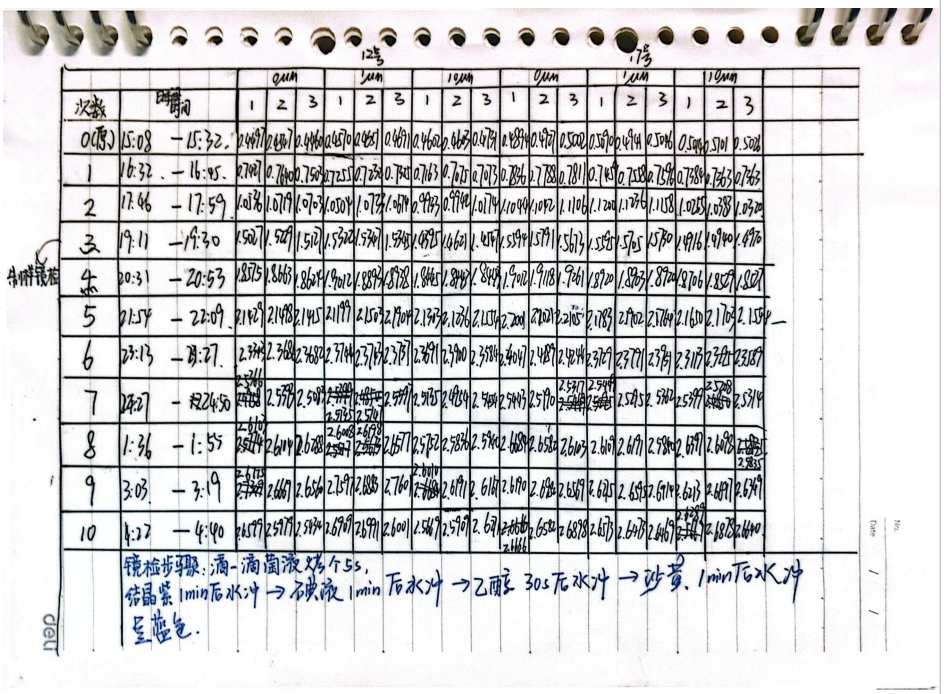
Electrophoretogram of the sixth colony PCR (the target band is 849 bp)

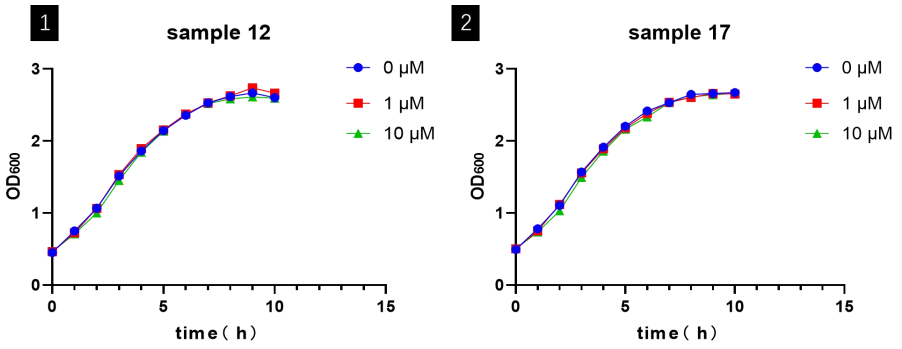
	1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24) 7.23  **Comparison diagrams of sequencing results (key humanized sites)** 1: alignment of sample 12 with the original sequence; 2: alignment of sample 17 with the original sequence; 3: alignment of sample 12 with the expected sequence; 4: alignment of sample 17 with the expected sequence
Explanation	SD-Ura screening only indicates that the *URA3* marker successfully entered the yeast cells; it cannot prove that the donor fragment was site-specifically integrated into the *GPA1* gene locus. In the earlier stage, multiple rounds of full-length primer amplification all failed; experimental condition issues such as marker interference and failure of the lysis solution were successively ruled out, and it was speculated that the original full-length primers had poor amplification efficiency for the recombinant strain. In the third round, the use of local fragment primers was able to amplify bands, indicating that fragment introduction had indeed occurred in some strains, but the full-length primers were difficult to detect it. Therefore, site-spanning specific primers were redesigned, with the upstream primer anchored outside the genomic homology arm and the downstream primer covering the humanized engineering region. Relying on this set of primers combined with sequencing screening, two yeast strains, 12 and 17, with completely correct genotypes for *GPA1* C-terminal humanization engineering were finally obtained.
Next plan	The reporter plasmid carrying the *yEGFP* fluorescent gene was transformed into positive strains 12 and 17.

No 12 Date：2026.07.29- 08.01 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Transformation of the *yEGFP* fluorescent plasmid into *GPA1* humanized positive strains and colony PCR verification
Note	Protocol 12（Yeast Transformation）、Protocol 13（Colony PCR for Saccharomyces cerevisia）、Protocol 5（Agarose Gel Electrophoresis）

Result	8.1  **Transformation plate images** 1: blank control; 2: transformed with the fluorescent plasmid 8.1  **Colony PCR electrophoretogram (the target bands are 1152 bp and 849 bp, respectively)** 1: 2000 bp marker; 2–5: verification of the fluorescent plasmid; 6–9: verification of the humanized sequence; 10: 2000 bp marker
Explanation	Through multiple rounds of PCR screening and sequencing identification in the earlier stage, strains 12 and 17 were confirmed as yeast engineered strains with completely correct site-specific integration for C-terminal humanization of *GPA1*. In this experiment, on that basis, the *yEGFP* fluorescent reporter plasmid was transformed in, and colony PCR was used for rapid verification of plasmid introduction. The results confirmed that the fluorescent plasmid was successfully introduced into the target strains, yielding dually verified engineered yeast possessing both the *GPA1* humanized background and the *yEGFP* fluorescent reporter system, thereby providing qualified experimental material for subsequent functional detection of fluorescent signals.
Next plan	Microscopic morphological examination, growth curve measurement, and functional verification of fluorescence signal intensity were carried out on the constructed fluorescent engineered strains.

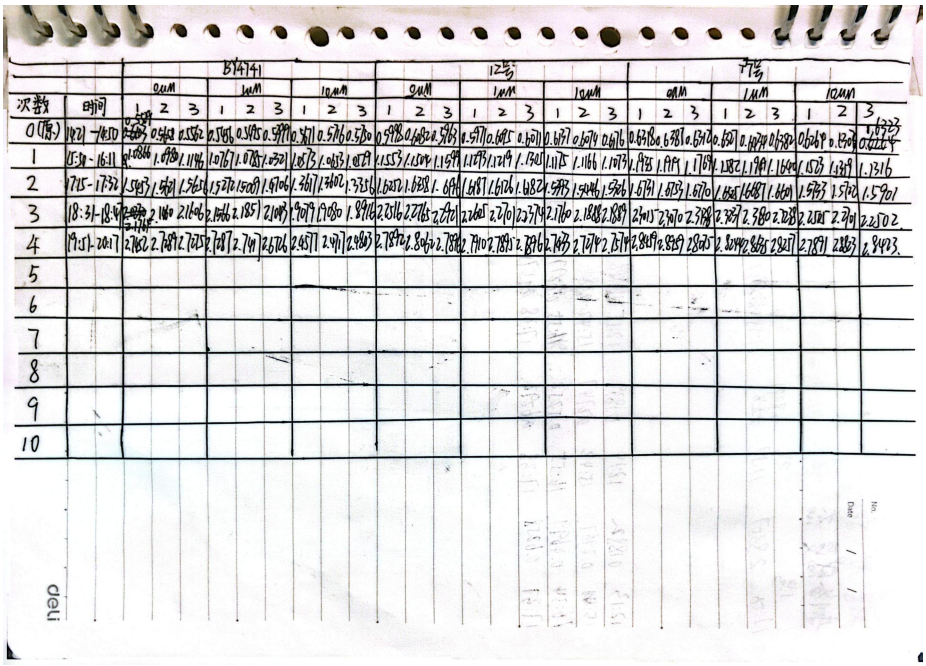
Experiment	Growth curve verification of the *GPA1* humanized yeast strains
Note	Protocol 14(Yeast Growth Curve Measurement)
Result	7.29  **Original record of the first growth curve measurement** 8.2  **Original record of the second growth curve measurement**



Growth curves obtained from the second measurement

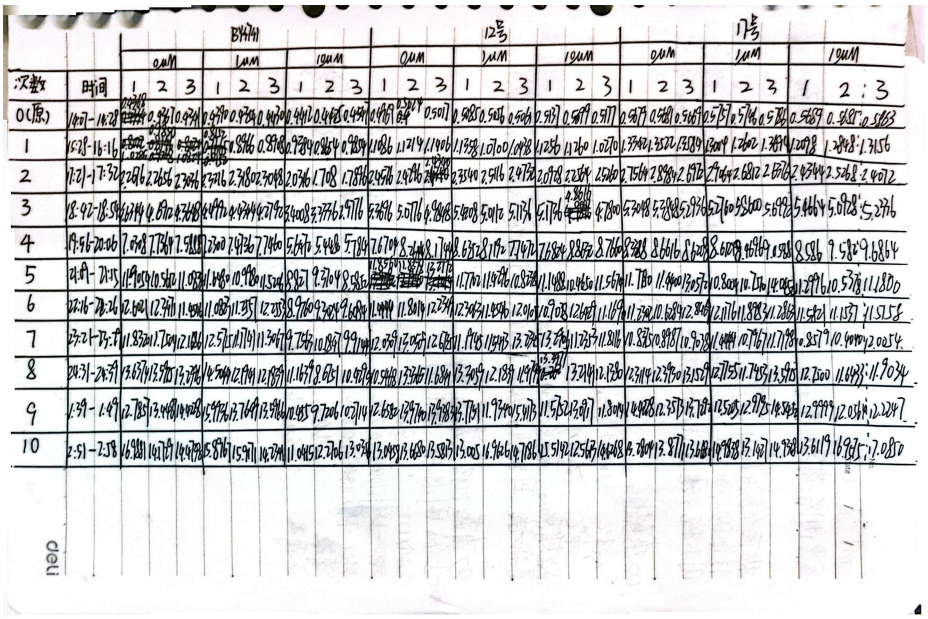
1: growth curve of sample 12; 2: growth curve of sample 17

8.8

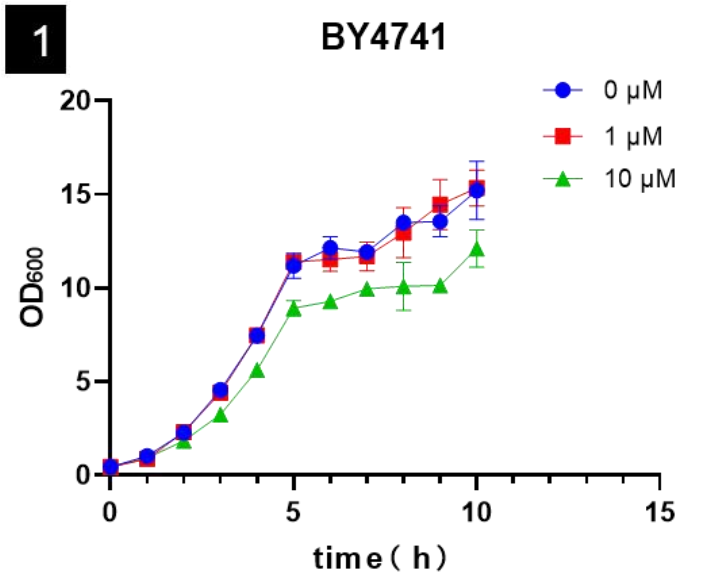


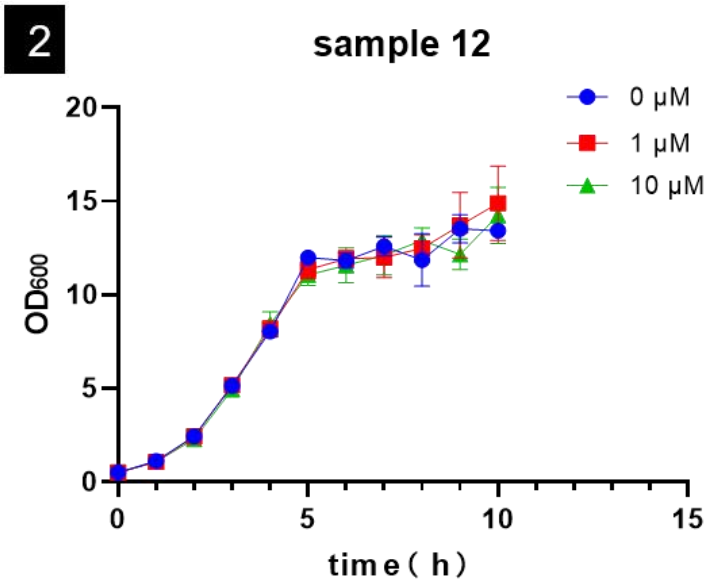
Original record of the third growth curve measurement

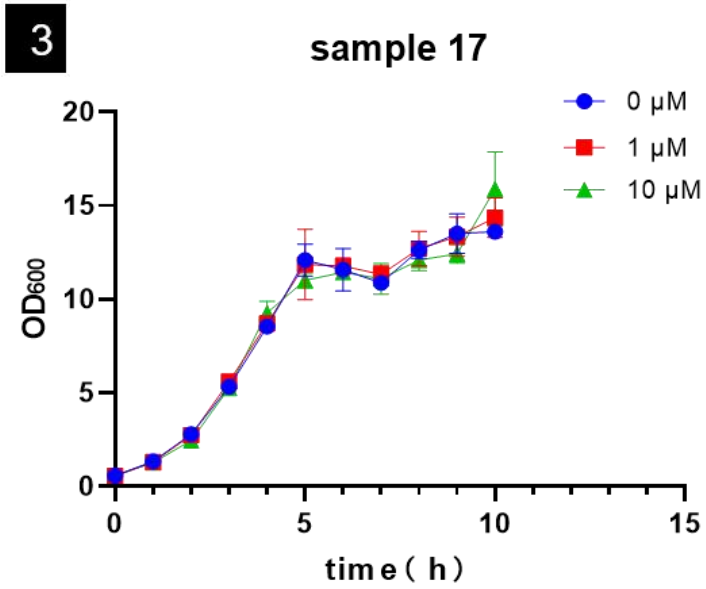
8.10



Original record of the fourth growth curve measurement


1
BY4741
0 μM
1 μM
10 μM
OD600
time (h)

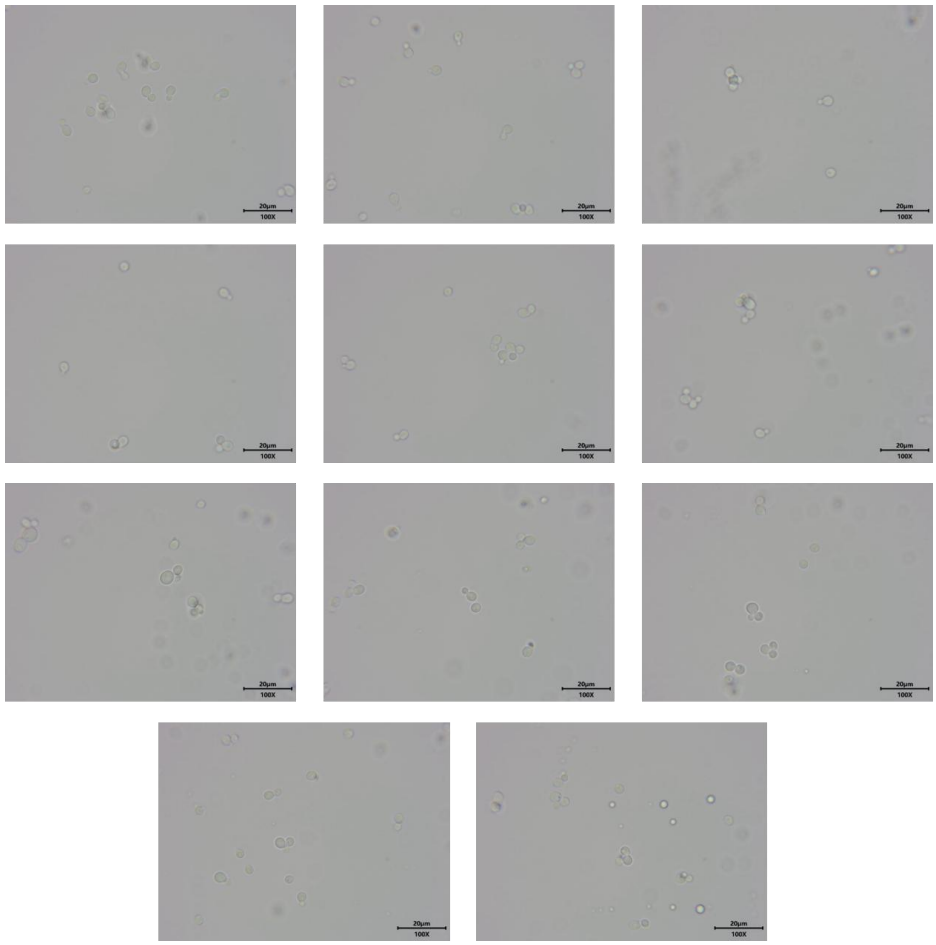

2
sample 12
0 μM
1 μM
10 μM
OD600
time (h)


3
sample 17
0 μM
1 μM
10 μM
OD600
time (h)

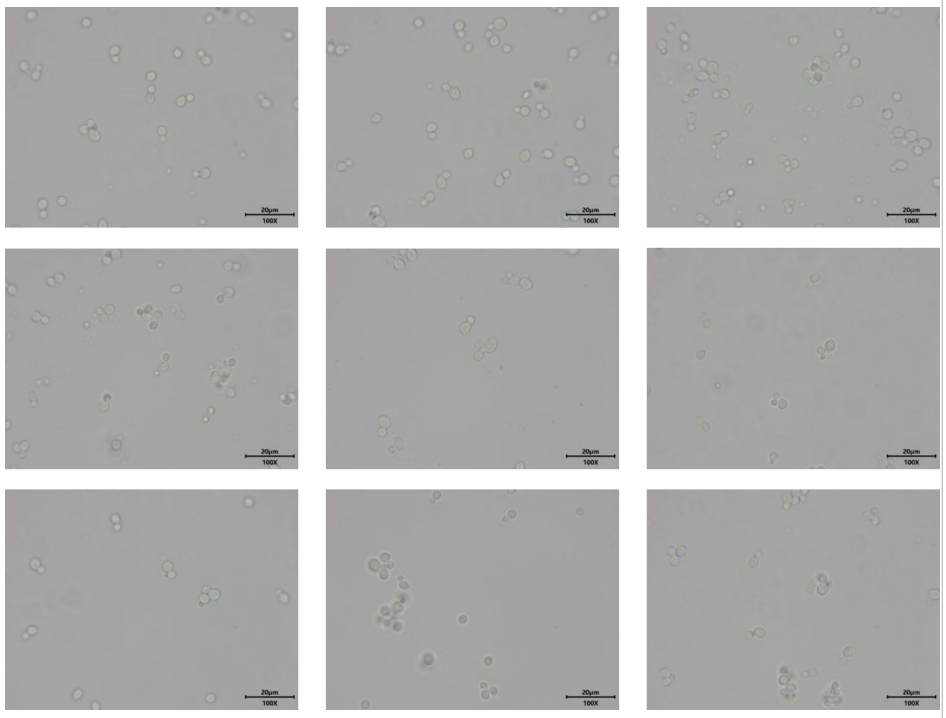
	Growth curves obtained from the fourth measurement 1: growth curve of *BY4741*; 2: growth curve of sample 12; 3: growth curve of sample 17
Explanation	This experiment used only the yeast strain itself, without introducing the *yEGFP* fluorescent plasmid, to evaluate the effects of C-terminal humanization editing of *GPA1* on yeast proliferation and on growth phenotypes under induction conditions. A total of four exploratory experiments were carried out: the first was interrupted due to contamination under YPD culture conditions; the second lacked a wild-type *BY4741* control, resulting in insufficient experimental completeness; in the third, the strains grew too rapidly, and the OD exceeded the linear detection range of the instrument, rendering the data invalid. By summarizing the problems from the first three experiments, dilution was added in the fourth experiment: when OD_{600} was greater than 0.6, the samples were diluted before measurement to ensure that the readings fell within the reliable range of the spectrophotometer, while wild-type *BY4741* was retained as a control. Phenotypic results showed that after high-concentration induction, the growth of wild-type *BY4741* was significantly inhibited, whereas the growth of the humanized strains 12 and 17 was unaffected, consistent with expectations. By comparing the growth rates and maximum OD of the wild-type strain with those of the humanized strains 12 and 17, it was confirmed that the gene-edited strains possessed the expected phenotype, providing a foundational background for subsequent microscopic morphological observation and fluorescence function experiments.
Next plan	Microscopic morphological observation of yeast cells was carried out.

No 14 Date：2026.08.01-08.02 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

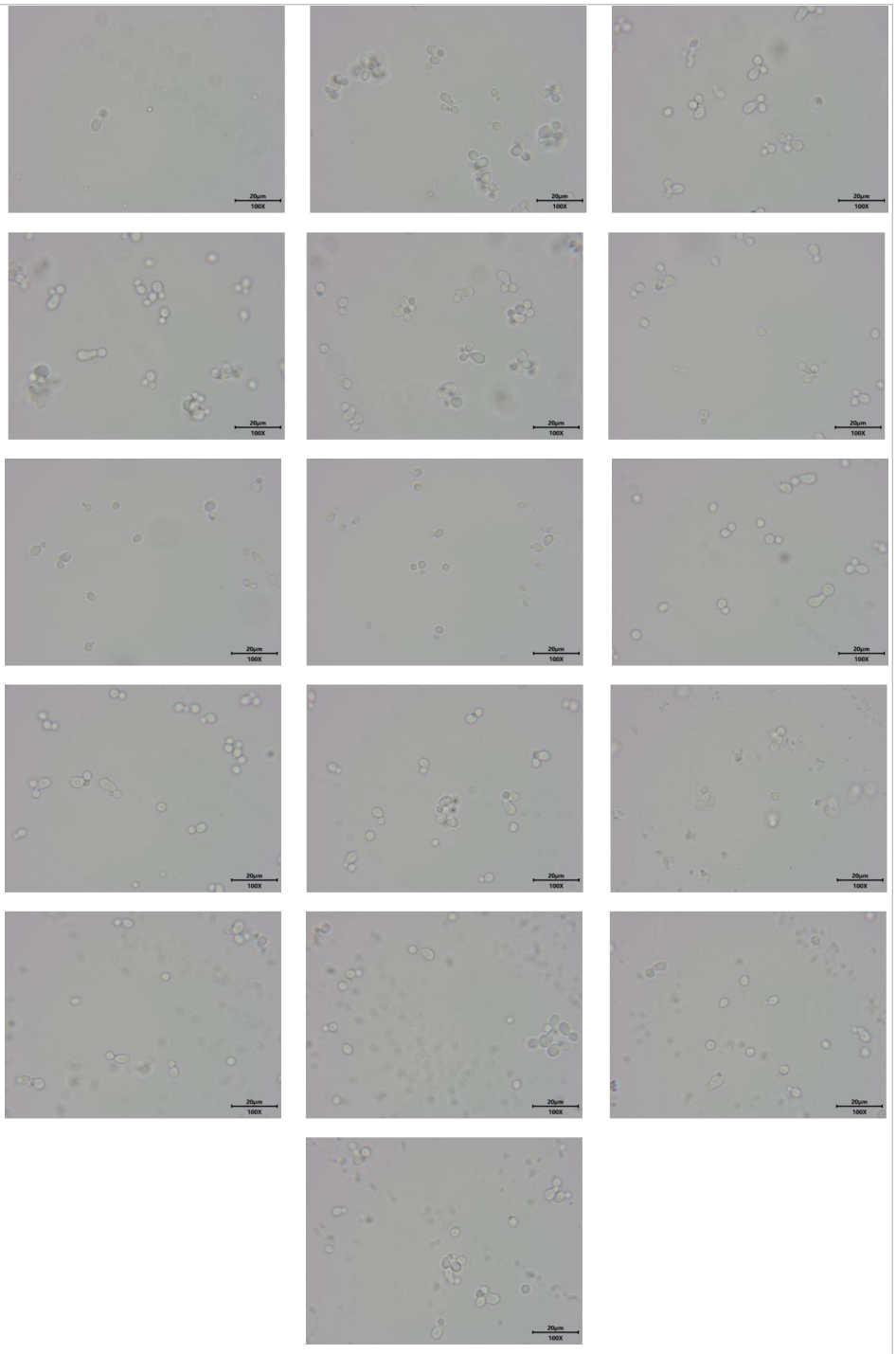
Experiment	Microscopic morphological observation of the *GPA1* humanized yeast strains
Note	Protocol 15(Microscopic Observation of Yeast Morphology)
Result	8.2 .



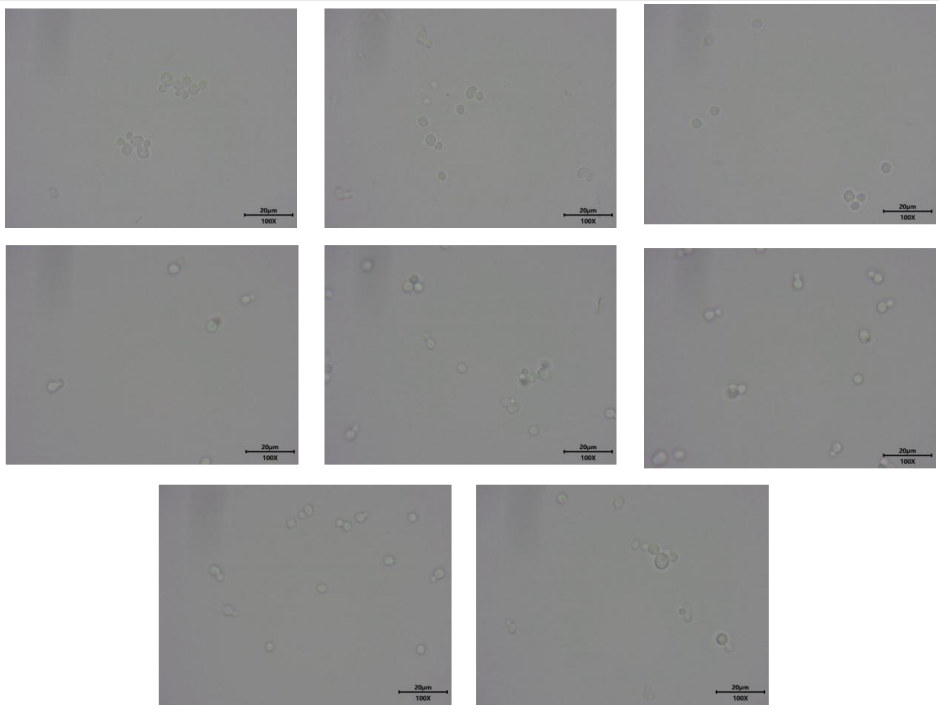
Microscopic image of sample 12 at OD600 of 0.4 under 0 µM α-factor



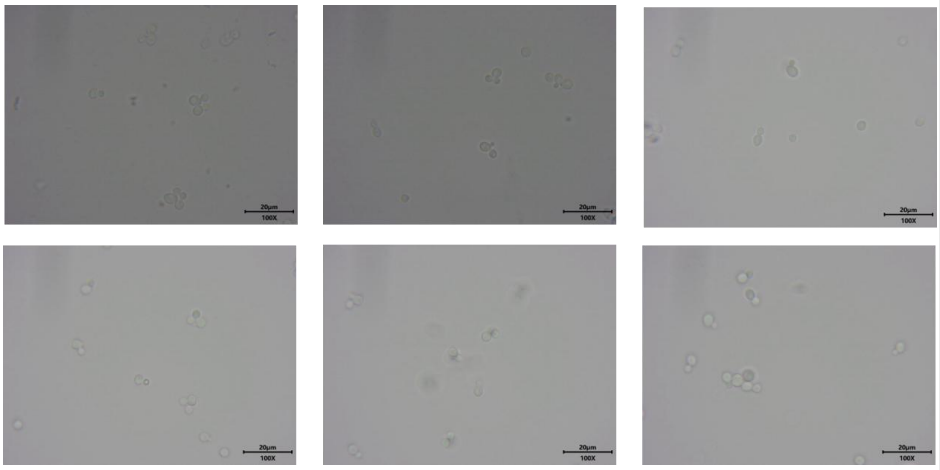
Microscopic image of sample 12 at OD600 of 0.4 under 1 µM α-factor



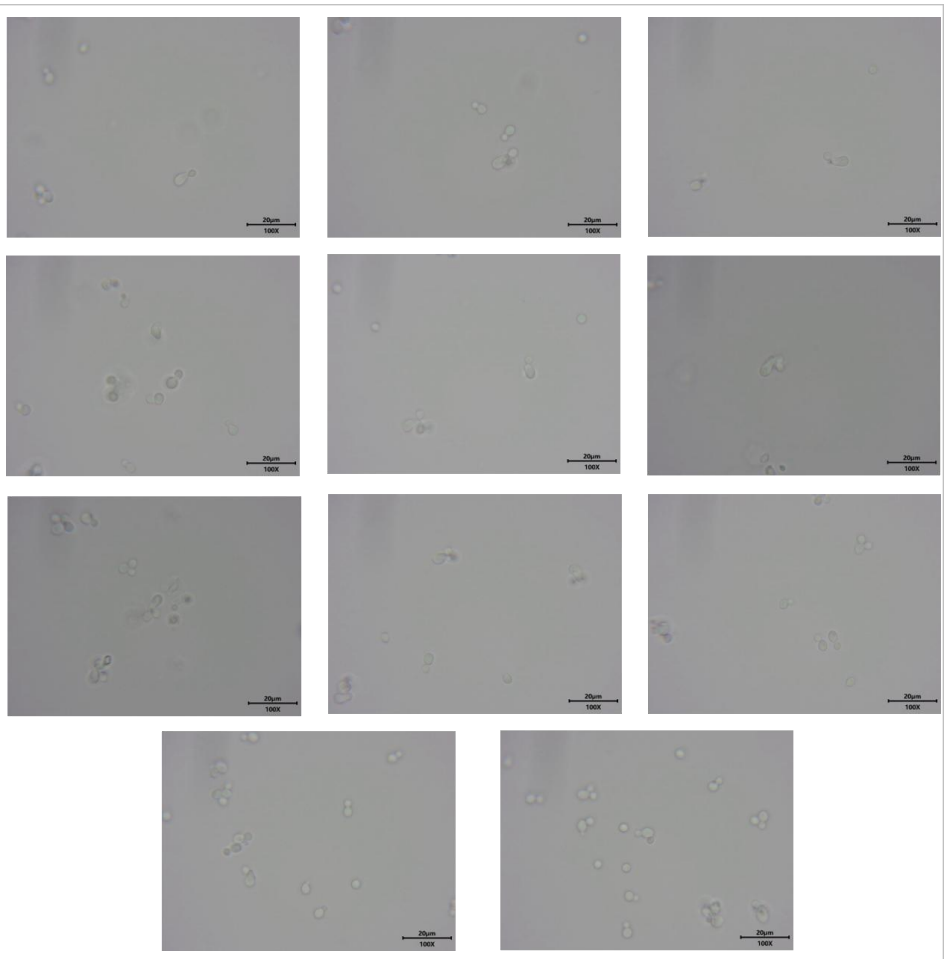
Microscopic image of sample 12 at OD600 of 0.4 under 10 µM α-factor



Microscopic image of sample 17 at OD600 of 0.4 under 0 µM α-factor



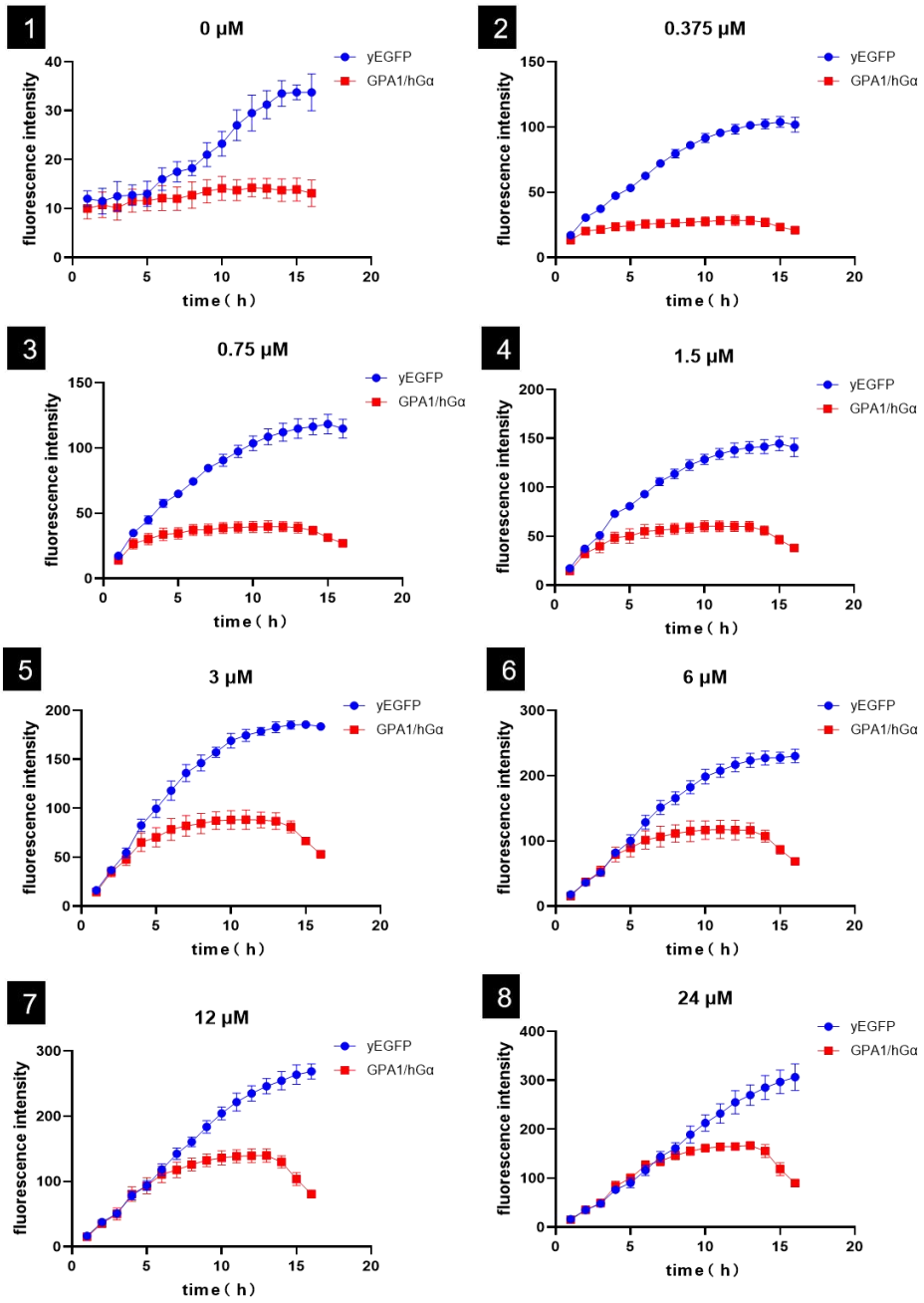
Microscopic image of sample 17 at OD600 of 0.4 under 1 µM α-factor

	 **Microscopic image of sample 17 at OD600 of 0.4 under 10 μM α-factor**
Explanation	On the basis of completing the growth curve measurement, the individual cell morphology was visually observed under a microscope to evaluate whether C-terminal humanization engineering of *GPA1* would cause cell morphology defects. Shmoo is a polarized protrusion phenotype produced by yeast in response to mating pheromone. Combining the observation results under treatments with different induction concentrations: high-concentration induction could cause the wild-type *BY4741* to produce the shmoo phenotype, whereas the humanized strains 12 and 17 showed no significant decrease in cell number and did not exhibit obvious shmoo mating protrusions. The microscopic morphological results and the growth phenotypes obtained from the growth curves corroborate each other, further demonstrating that the engineered strains were in good physiological condition and could be used for subsequent fluorescence signal detection experiments.
Next plan	Fluorescence signal intensity detection of yeast was carried out.

No 14 Date：2026.08.04- 08.05 Experimenters: Yuelin Zheng、 Chengxi Luo Recorder: Yuelin Zheng

Experiment	Fluorescence signal intensity detection of *GPA1* humanized yeast carrying the *yEGFP* reporter plasmid

Note	Protocol 16(Detection of Yeast Fluorescence Signal After Treatment)
Result	8.5



Fluorescence signal intensity curves of *GPA1* humanized yeast carrying the *yEGFP* reporter plasmid

1: fluorescence intensity change curve under 0 μM α-factor induction; 2: fluorescence intensity change curve under 0.375 μM α-factor induction; 3: fluorescence intensity change curve under 0.75 μM α-factor induction; 4: fluorescence intensity change curve under 1.5 μM α-factor induction; 5: fluorescence intensity change curve under 3 μM α-factor induction; 6: fluorescence intensity change curve under 6 μM α-factor induction; 7: fluorescence intensity change curve under 12 μM α-factor induction; 8:

	fluorescence intensity change curve under 24 μM α-factor induction
Explanation	In the earlier stage, genotypic sequencing verification of the strains, growth curve measurement, and microscopic morphological observation of the cells had already been completed, confirming that the humanized strains 12 and 17 were in normal physiological condition. In this experiment, with the aid of the *yEGFP* fluorescent reporter system, the activation level of the intracellular reporter gene under different induction conditions was detected, thereby reflecting the activation response of the G protein signaling pathway. Fluorescence values were normalized to OD_{600} to eliminate interference caused by differences in cell concentration. The fluorescence kinetics results indicated that C-terminal humanization engineering of *GPA1* altered the signal output characteristics of the yeast G protein pathway: the engineered strains possessed a higher peak fluorescence response and an extended time to peak, reflecting a unique signal response kinetics mediated by the humanized Gα subunit, thereby completing the verification of the major phenotypic functions of the engineered strains.
Next plan	All experimental data from the entire set of strain construction and phenotypic testing in this study were organized and compiled, and strain preservation in 80 °C glycerol was carried out according to Protocol 2.